Adaptive Mask Sampling and Manifold to Euclidean Subspace Learning With Distance Covariance Representation for Hyperspectral Image Classification

Mingsong Li, Wei Li, *Senior Member, IEEE*, Yikun Liu, Yuwen Huang, and Gongping Yang

***Abstract*— For the abundant spectral and spatial information recorded in hyperspectral images (HSIs), fully exploring spectral–spatial relationships has attracted widespread attention in the HSI classification (HSIC) community. However, there are still some intractable obstructs. For one thing, in the patch-based processing pattern, some spatial neighbor pixels are often inconsistent with the central pixel in land-cover class. For another thing, linear and nonlinear correlations between different spectral bands are vital yet tough for representing and excavating. To overcome these mentioned issues, an adaptive mask sampling and manifold to the Euclidean subspace learning (AMS-M2ESL) framework is proposed for HSIC. Specifically, an adaptive mask-based intrapatch sampling (AMIPS) module is first formulated for intrapatch sampling in an adaptive mask manner based on central spectral vector-oriented spatial relationships. Subsequently, based on the distance covariance descriptor, a dual-channel distance covariance representation (DC-DCR) module is proposed for modeling unified spectral–spatial feature representations and exploring spectral–spatial relationships, especially linear and nonlinear interdependence in the spectral domain. Furthermore, considering that the distance covariance matrix lies on the symmetric positive definite (SPD) manifold, we implement an M2ESL module respecting the Riemannian geometry of the SPD manifold for high-level spectral–spatial feature learning. Additionally, we introduce an approximate matrix square-root (ASQRT) layer for efficient Euclidean subspace projection. Extensive experimental results on three popular HSI datasets with limited training samples demonstrate the superior performance of the proposed method compared with other state-of-the-art methods. The source code is available at https://github.com/lms-07/AMS-M2ESL.**

***Index Terms*— Adaptive mask sampling (AMS), distance covariance representation (DCR), hyperspectral image classification (HSIC), manifold to Euclidean subspace learning (M2ESL).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral signatures from more than hundreds of contiguous narrow spectral bands and plentiful spatial information on land covers. Thus, HSIs have played a substantial role in many fields, for example, precision agriculture, mineral exploitation, and environment management [1], [2], [3]. As a fundamental yet challenging task of HSI processing and analysis, HSI classification (HSIC) aims to assign a unique land-cover label for each HSI pixel based on the set of given land-cover classes.

With decades of development, both traditional feature engineering methods and deep-learning methods make great progress on the HSIC task. The former ones either directly identify and select the set of the most representative spectral bands, for example, constrained band selection (CBS) [4] and clustering-based band selection [5], or learn a new set of representative features via linear or nonlinear transformations, for example, principal component analysis (PCA) [6], minimum noise fraction (MNF) [7], and locally linear embedding (LLE) [8]. Different from the shallow manner of feature engineering methods, the latter ones extract more abstract and discriminative features of HSIs in a hierarchical manner, including stacked autoencoders (SAEs) [9], convolutional neural networks (CNNs) [10], [11], [12], [13], graph neural networks (GNNs) [14], [15], [16], recurrent neural networks (RNNs) [17], [18], generative adversarial networks (GANs) [19], and transformers [20], [21].

Although deep-learning-based HSIC methods have strong feature representation ability and promising classification performance, there are still some intractable obstructs toward the sufficient mining of spectral–spatial relationships.

A. Complicated Spatial Land-Cover Distribution

For one thing, HSIs are captured by imaging spectrometers mounted on aircraft or satellites, thus they are capable of recording complicated land-cover distribution. For another thing, due to the tradeoff between spectral and spatial resolution in the hyperspectral imaging process, spatial resolution

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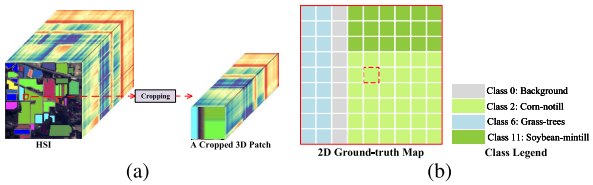


Fig. 1. (a) Patch-based processing pattern of the HSIC task, where the image instance is from the IP dataset and the patch size is set as 9×9 (referring to Section IV-B). (b) Detailed 2-D ground-truth map for all the pixels of the cropped 3-D patch instance, the central pixel is marked red dotted box. The land-cover label of the cropped 3-D patch instance and the central pixel is both "corn-notill," but part of the neighbor pixels belong to other land-cover classes, that is, "grass-trees" and "Soybean-mintill."

usually becomes a victim [22], [23] and is lower than 1 m/pixel magnitude. In the common patch-based processing pattern of HSIC, the complicated land-cover distribution and the lower spatial resolution make a small patch cover a wide observed scene and span multiple land-cover classes, which usually exacerbates the inconsistency in the land-cover class between the central pixel and the neighbor pixels of the patch.

In detail, to conveniently aggregate information from spectral and spatial domains, the patch-based processing pattern is the most popular image processing and representing pattern for the HSIC task. As shown in Fig. 1(a), a 3-D spectral–spatial patch cropped from the original HSI is the input unit of an HSIC model, which represents its central pixel and is coherent with its central pixel in the land-cover label. Based on the assumption that the neighbor pixels are likely to belong to the same land-cover category [24], [25], these neighbor pixels play auxiliary roles in land-cover classification processing. However, as shown in Fig. 1(b), due to the complicated land-cover distribution and the low spatial resolution, the central pixel and its neighbor pixels may be inconsistent in the land-cover category, which inevitably produces a negative impact on representing and exploiting spectral–spatial relationships and finally decreases the classification performance.

To alleviate this problem, Zhao et al. [13] designed a novel superpixel-guided deformable convolution network (SGDCN), which is able to make the shape of the convolution kernel fit with the shape of the land-cover object. For the complicated land-cover distribution, Sun et al. [25] proposed a fine label style to label all pixels of the HSI patch and an HSI patch generation method to generate plentiful fine labeled patches. As for the attention mechanism-related solutions, based on the bilinear fashion [26], Xue et al. [27] proposed an attention-based second-order pooling operator (A-SOP) employing cosine distance to measure the similarity between each neighboring pixel and the corresponding central pixel. Subsequently, Zhang et al. [28] presented a spectral–spatial self-attention network (SSSAN) in spectral–spatial dual-branch manner, where a spatial self-attention module was designed to strengthen the patch-based long-range contextual information related to the center pixel.

B. Complex Spectral Interdependence

With rich spectral signatures across more than hundreds of spectral bands, it is essential to explore the interdependence of different bands. Besides the common linear interrelationship, due to multiple complicated scattering mechanisms, heterogeneous composition of pixels, interclass similarity, and intraclass variability [3], [29], [30], [31], [32], [33], [34], [35], [36], nonlinear interdependence in the spectral domain actually exists, yet is tricky. Therefore, many advanced subspace learning and manifold learning methods were developed for the exploitation of spectral relationships, for example, improved isometric mapping (ISOMAP) [29], spherical stochastic process embedding framework [30], joint and progressive learning strategy (J-Paly) [32], and manifold learning-based semisupervised neural network (MSS-Net) [33].

However, the above-mentioned methods only implicitly explore spectral interdependence without explicit representation. To this end, covariance matrix representation, the second-order statistical descriptor, was introduced to fully characterize spectral correlations and spatial contextual information of each local neighbor region [34]. Then He et al. [35] made an attempt to directly assembled multiscale covariance maps (MCMs) with the 2-D-CNN. Recently, Zhang et al. [36] introduced distance covariance matrix [37], [38] and proposed a superpixel-based Brownian descriptor (SBD) method to explicitly model the linear and nonlinear spectral interdependence and spatial structure information of each homogeneous superpixel block.

Furthermore, covariance-based methods [34], [35], [36] have explored an elegant spectral–spatial representation of each 3-D patch or homogeneous neighbor region. But there is still potential for sufficient exploitation of spectral–spatial relationships. First, both covariance matrix [34], [35] and distance covariance matrix [36] are symmetric positive definite (SPD) matrices, which underlie on SPD manifold, that is, a type of nonlinear Riemannian manifold [39], [40], [41], [42] and require non-Euclidean geometry metrics, for example, Log-Euclidean (Log-E) metric [39] used in [34] and [36]. On the contrary, MCM-CNN [35] ignored the Riemannian geometry of the SPD manifold of MCM and directly applied MCM for the CNN model. Second, the classic covariance matrix only measures the degree of linear correlation among spectral bands, and the distance covariance matrix is able to capture nonlinear and nonmonotone interdependence besides linear correlation [36], [38]. Finally, considering the limited feature extraction ability of the shallow manner, we claim that the suitable and reasonable deep-learning manner respecting the Riemannian geometry of the SPD manifold would sufficiently utilize and excavate the spectral–spatial representation of the covariance matrix in the HSIC task.

C. Limited Labeled Samples

Compared with classic natural images, the imaging and acquisition of HSIs are more difficult since they usually rely on imaging spectrometers mounted on aerospace platforms. As discussed in problem 2) (Section I-B), the imaging process is susceptible to the object scatting conditions and the introduced noises. Meanwhile, the labeling of HSIs greatly depends on domain knowledge and is time-consuming and labor-intensive. Therefore, only utilizing limited labeled samples to improve the ability of spectral–spatial feature extraction becomes a crucial issue in the HSIC community.

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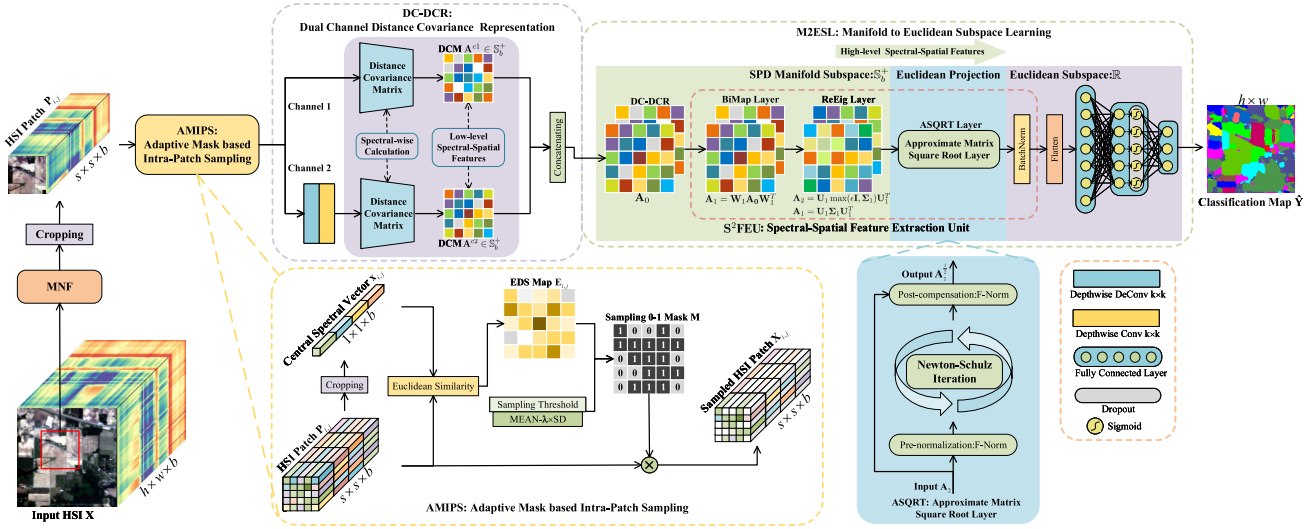


Fig. 2. Flowchart of the proposed AMS-M2ESL framework, which mainly consists of an AMIPS module employing an adaptive spatial self-similarity mask mechanism for intrapatch sampling, a DC-DCR module for modeling unified spectral–spatial feature representations, and an M2ESL module to mine spectral–spatial DCR in two feature subspaces, that is, SPD manifold subspace and Euclidean subspace. The Euclidean projection between the two feature subspaces is an ASQRT layer.

To cope with the three above-mentioned problems, we propose an adaptive mask sampling and manifold to the Euclidean subspace learning (AMS-M2ESL) framework with limited labeled samples for HSIC. Specifically, the raw HSI first reduced dimensionality and discarded noises through MNF. Then as a new insight to alleviate the above-discussed land-cover inconsistency problem, we elaborate an adaptive mask-based intrapatch sampling (AMIPS) module, which produces an adaptive mask for intrapatch sampling based on central spectral vector-oriented spatial relationships. Moreover, based on distance covariance descriptor [37], [38], we design a dual-channel distance covariance representation (DC-DCR) module to model unified spectral–spatial feature representations and explore spectral–spatial relationships, especially linear and nonlinear interdependence in the spectral domain. Furthermore, respecting the Riemannian geometry of the SPD manifold, we implement an M2ESL module to sufficiently mine spectral–spatial DCR in two feature subspaces. Additionally, we introduce a graphics processing unit (GPU)-friendly Newton–Schulz iteration [43], [44] method to efficiently compute matrix square root (SQRT) for Euclidean subspace projection in the M2ESL module. With the intrapatch sampling of the AMIPS module, the low-level spectral–spatial feature representation of the DC-DCR module, and the high-level spectral–spatial feature extraction of the M2ESL module, the proposed AMS-M2ESL framework is capable of extracting discriminant spectral–spatial feature and achieving outstanding classification performance only utilizing limited labeled samples on three popular datasets.

To the best of our knowledge, it is the first time to mine DCR in the deep SPD manifold learning manner respecting Riemannian geometry of SPD manifold in the HSIC task. Fig. 2 illustrates the proposed framework, in which the Indian Pines (IP) dataset is taken into account. The main contributions of this article are summarized as follows.

1) Based on central spectral vector-oriented spatial relationships of each patch, we formulate an AMIPS module for intrapatch sampling in an adaptive mask matrix manner, which provides a new insight to alleviate the complicated spatial land-cover inconsistency problem.

2) Based on distance covariance descriptor, we design a DC-DCR module to model unified spectral–spatial feature representations and explore spectral–spatial relationships, especially the linear and nonlinear spectral interdependence.

3) Respecting the Riemannian geometry of the SPD manifold, we implement an M2ESL module to sufficiently mine spectral–spatial DCR in two feature subspaces. Additionally, we introduce an approximate matrix SQRT (ASQRT) layer for efficient Euclidean subspace projection. The powerful high-level feature extraction ability of the AMS-M2ESL module promotes the classification performance with limited labeled samples.

The remainder of this article is organized as follows. Section II briefly overviews related backgrounds about distance covariance descriptor and deep SPD manifold learning. Section III introduces the proposed AMS-M2ESL framework in detail. Experimental results and corresponding analyses are reported in Section IV. Finally, Section V summarizes this article.

II. Background

In this section, related backgrounds and basic concepts about the distance covariance descriptor and deep SPD manifold learning are briefly introduced.

A. Distance Covariance Descriptor

Formally, let $\mathbf{x} \in \mathbb{R}^p$ and $\mathbf{y} \in \mathbb{R}^q$ be random vectors of dimension p and q, respectively. $\mathrm{Cov}(\mathbf{x}, \mathbf{y})$ and $\mathcal{V}(\mathbf{x}, \mathbf{y})$ denote the corresponding classic covariance and distance covariance of the observation pair $(\mathbf{x}, \mathbf{y})$, respectively.

The covariance descriptor as a popular statistical representation is able to encode the correlation of samples, which has been widely employed and developed in many vision tasks,

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for example, image set classification [42], [45], second-order pooling [26], [43], [44], and HSIC [34], [35]. The limits of the classic covariance descriptor are simply as follows: 1) classic covariance is mainly sensitive to linear dependence between two random variables, and nonlinear or nonmonotone dependence may exist; and 2) $\mathrm{Cov}(\mathbf{x}, \mathbf{y}) = 0$ only imply $\mathbf{x}$ and $\mathbf{y}$ are uncorrelated, in general.

Thus, distance covariance descriptor was introduced as a new approach to measuring linear and nonlinear dependence random vectors in arbitrary dimension [37], [38]. Some vital properties of distance covariance are as follows: 1) $0 \leq \mathcal{V}(\mathbf{x}, \mathbf{y}) \leq 1$ and $\mathcal{V}(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x}$ and $\mathbf{y}$ are independent and 2) $\mathcal{V}(\mathbf{x}, \mathbf{y})$ can characterize linear and nonlinear dependency between $\mathbf{x}$ and $\mathbf{y}$.

Despite distance covariance being a powerful modeling descriptor, there has been little related work in vision tasks. Distance covariance descriptor was introduced as a new image region descriptor to characterize feature intercorrelations for pedestrian detection [46]. Recently, following the second-order pooling manner, a deep Brownian distance covariance (DeepBDC) method was formulated as a spatial pooling layer and further embedded into two different few shot classification frameworks [47]. As discussed in Section I, the SBD method was proposed to construct discriminative spectral–spatial feature representation for HSIC [36].

B. Deep SPD Manifold Learning

1) SPD Manifold: An $n \times n$ symmetric real matrix $\mathbf{X}$ satisfies positive definite property iff $\mathbf{v}^T\mathbf{X}\mathbf{v} > 0$ for all nonzero vector $\mathbf{v} \in \mathbb{R}^n$, and SPD manifold denoted as $\mathbb{S}_n^+$ is spanned by a set of $n \times n$ SPD matrices.

2) Basic Layers of SPD Matrix Network: Recently, analogous to standard fully connected networks, Huang et al. [40] proposed an SPD matrix network (SPDNet) consisting of some basic SPD layers to perform deep-learning respecting the SPD manifold. Let $\mathbf{X}_{k-1} \in \mathbb{S}_n^+$ be the input SPD matrix to k^{th} layer, and $\mathbf{U}_{k-1}$ and $\mathbf{\Sigma}_{k-1}$ be the orthonormal matrix and singular-value matrix, respectively, produced via eigenvalue decomposition (EIG) of $\mathbf{X}_{k-1}$, that is, $\mathbf{X}_{k-1} = \mathbf{U}_{k-1}\mathbf{\Sigma}_{k-1}\mathbf{U}_{k-1}^T$. The basic SPD layers are defined as follows.

1) BiMap layer: This layer generates a more compact and discriminative SPD matrix with a transformation matrix $\mathbf{W}_k$ by a bilinear mapping as $\mathbf{X}_k = \mathbf{W}_k\mathbf{X}_{k-1}\mathbf{W}_k^T$. To ensure the output $\mathbf{X}_k$ to be SPD, $\mathbf{W}_k$ is required to be row full-rank through a semiorthogonal constraint on it.
2) ReEig layer: This layer as a ReLU-like layer [48] introduces nonlinearity into SPDNet, which is defined as $\mathbf{X}_k = \mathbf{U}_{k-1}\max(\epsilon\mathbf{I}, \mathbf{\Sigma}_{k-1})\mathbf{U}_{k-1}^T$, where ϵ is a rectification threshold, and $\mathbf{I}$ is the identity matrix.
3) LogEig layer: Based on Log-E Riemannian metric [39], LogEig layer is designed to project the feature manifold to a Euclidean space via matrix logarithm as $\mathbf{X}_k = \mathbf{U}_{k-1}\log(\mathbf{\Sigma}_{k-1})\mathbf{U}_{k-1}^T$.

III. Proposed Method

Fig. 2 illustrates the flowchart of the proposed AMS-M2ESL framework, which mainly consists of three modules, that is, an AMIPS module for intrapatch sampling, a DC-DCR module for modeling unified spectral–spatial feature representation, and an M2ESL module for sufficiently mining spectral–spatial DCR in two feature subspaces. Detailed introductions about all three modules are presented in Sections III-A to III-C.

Given an HSI $\mathbf{X} \in \mathbb{R}^{h\times w\times b}$, HSIC task is assigning a land-cover category label $y_{i,j} = k, k = 1, \ldots, K$ for each pixel $\mathbf{x}_{i,j} \in \mathbb{R}^b$ in accordance with a couple of the labeled pixel set $\mathbf{X}_{\{l\}} = \{\mathbf{x}_1, \ldots, \mathbf{x}_l\}$ and label set $\mathbf{Y}_{\{l\}} == \{y_1, \ldots, y_l\}$, where h, w, b, K, and l represent the height of spatial dimension, the width of spatial dimension, the number of spectral bands, the number of classes, and the number of labeled pixels, respectively. $i = 1, \ldots, h$ and $j = 1, \ldots, w$ jointly represent the coordinate of a pixel in HSI $\mathbf{X}$. In the patch-based processing pattern of HSIC, the raw HSI $\mathbf{X}$ is cropped into a set of overlapping patches $\mathbf{P}$, and each patch $\mathbf{P}_{i,j} \in \mathbb{R}^{s\times s\times b}$ in $\mathbf{P}$ covering a $s \times s$ size spatial window and all the b spectral bands are centered at pixel $\mathbf{x}_{i,j}$, and the label of $\mathbf{P}_{i,j}$ is coherent with the label of $\mathbf{x}_{i,j}$.

In the proposed framework, before cropping the raw HSI $\mathbf{X}$, we first adopt band-wise de-mean and variance standardization to facilitate convergence and then employ MNF [7] to reduce computational cost and discard possible noises introduced in the imaging process.

A. Adaptive Mask-Based Intrapatch Sampling

As illustrated in Fig. 1, due to the complicated land-cover distribution and the low spatial resolution, a small patch is able to cover a large observed scene and span multiple land-cover classes in a patch, which usually amplifies the inconsistency in the land-cover class between the central pixel and neighbor pixels. Recently, deformable convolution-based method [13], fine label style-based method [25], and attention-based methods [27], [28] were proposed to alleviate this problem. In the proposed framework, we propose a new insight to formulate an AMIPS module to sample the neighbor pixels with high similarity to the central pixel in an adaptive mask manner.

Formally, let an HSI patch $\mathbf{P}_{i,j} \in \mathbb{R}^{s\times s\times b}$ be the module input. As shown in Fig. 2, the central spectral vector $\mathbf{x}_{i,j} \in \mathbb{R}^b$ is first selected through cropping the patch $\mathbf{P}_{i,j}$. Second, we calculate Euclidean distance similarity (SimED) between the central spectral vector $\mathbf{x}_{i,j}$ and the HSI patch $\mathbf{P}_{i,j}$ to produce SimED map $\mathbf{E}_{i,j} \in \mathbb{R}^{s\times s}$, which describes central spectral vector-oriented spatial relationships, that is,

$$\mathbf{E}_{i,j} = \mathrm{EDSim}\left(\mathbf{x}_{i,j}, \mathbf{P}_{i,j}\right). \tag{1}$$

Denote $e_{i',j'}$ as the SimED value located at (i', j') of $\mathbf{E}_{i,j}$, where $i' = 1, \ldots, s$ and $j' = 1, \ldots, s$, which can be formulated as

$$\begin{aligned} e_{i',j'} &= \mathrm{EDSimV}\left(\mathbf{x}_{i,j}, \mathbf{x}_{i',j'}\right) \\ &= \frac{1}{1+\parallel \mathbf{x}_{i,j} - \mathbf{x}_{i',j'} \parallel} \end{aligned} \tag{2}$$

where $\mathbf{x}_{i',j'}$ is the corresponding neighbor spectral vector coordinated at (i', j') of $\mathbf{P}_{i,j}$. Here, the similarity value has been normalized as the unified representation that the closer

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Algorithm 1 AMIPS

Input: An HSI patch $\mathbf{P}_{i,j} \in \mathbb{R}^{s\times s\times b}$, the sampling threshold hyperparameter λ.

Output: The sampled patch $\mathbf{X}_{i,j} \in \mathbb{R}^{s\times s\times b}$.

for *each input HSI patch* $\mathbf{P}_{i,j}$ **do**
- Crop the central spectral vector $\mathbf{x}_{i,j} \in \mathbb{R}^{b}$ from $\mathbf{P}_{i,j}$.
- Compute the Euclidean distance similarity map $\mathbf{E}_{i,j} \in \mathbb{R}^{s\times s}$ by Eqs. (1)-(2).
- Compute the mean m_s and standard deviation s_s of $\mathbf{E}_{i,j}$.
- Compute sampling threshold $t_s = m_s - \lambda \times s_s$.
- Compute the sampling 0-1 mask matrix $\mathbf{M} \in \mathbb{R}^{s\times s}$ by Eq. (3).
- Perform adaptive mask based intra-patch sampling by Eq. (4).
- Return the sampled patch $\mathbf{X}_{i,j} \in \mathbb{R}^{s\times s\times b}$.

end

the similarity value is to 1, the higher the similarity between the coupled vectors, that is, $\mathbf{x}_{i,j}$ and $\mathbf{x}_{i',j'}$.

Moreover, for adaptively selecting high confidence neighbor pixels, based on the mean m_s and standard deviation s_s of the similarity map $\mathbf{E}_{i,j}$, we set sampling threshold t_s as $t_s = m_s - \lambda \times s_s$, where λ is a hyperparameter discussed in Section IV-B. In fact, as long as λ is determined, the sampling threshold t_s adaptively depends on the central spectral vector-oriented spatial relationships of each patch, and this kind of threshold setting manner is popular in vision tasks, for example, [49], [50].

Furthermore, based on the sampling threshold t_s, the sampling 0-1 mask matrix $\mathbf{M} \in \mathbb{R}^{s\times s}$ can be further formulated as

$$m_{i',j'} = \begin{cases} 1, & e_{i',j'} \geq t_s \\ 0, & e_{i',j'} \leq t_s \end{cases} \tag{3}$$

where $m_{i',j'}$ is the value of $\mathbf{M}$ at location (i', j'). Finally, based on the adaptive sampling 0-1 mask matrix $\mathbf{M}$, the sampled patch $\mathbf{X}_{i,j} \in \mathbb{R}^{s\times s\times b}$ can be produced by

$$\mathbf{X}_{i,j} = \mathbf{M} \otimes \mathbf{P}_{i,j} \tag{4}$$

where $\otimes$ represents element-wise multiplication along the spectral dimension. The complete process of the proposed AMIPS module is illustrated in Algorithm 1.

B. Dual-Channel Distance Covariance Representation

For HSIs, characterizing linear and nonlinear interdependence in the spectral domain is significant yet tricky. Based on the elegant and powerful distance covariance descriptor [37], [38], the proposed DC-DCR module models unified spectral–spatial feature representations and explores spectral–spatial relationships, especially linear and nonlinear interdependence in the spectral domain.

As shown in Fig. 2, in formal, the DC-DCR module takes the sampled patch $\mathbf{X}_{i,j} \in \mathbb{R}^{s\times s\times b}$ as its module input. To obtain more sufficient spectral–spatial feature representation, a parallel dual channel structure is adopted, where

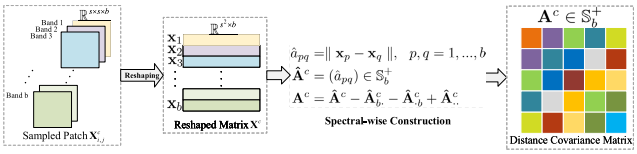


Fig. 3. Spectral-wise construction process of the distance covariance matrix.

channel 1 maintains the original spectral–spatial information of $\mathbf{X}_{i,j}$ and channel 2 extracts more abstract spectral–spatial information of $\mathbf{X}_{i,j}$. In detail

$$\begin{aligned} \mathbf{X}_{i,j}^{c1} &= \mathbf{X}_{i,j} \\ \mathbf{X}_{i,j}^{c2} &= \text{DW-DConv}_{k\times k}\big(\text{DW-Conv}_{k\times k}\big(\mathbf{X}_{i,j}\big)\big) \end{aligned} \tag{5}$$

where $\mathbf{X}_{i,j}^{c1} \in \mathbb{R}^{s\times s\times b}$ and $\mathbf{X}_{i,j}^{c2} \in \mathbb{R}^{s\times s\times b}$ are the inputs to construct distance covariance matrix in channels 1 and 2, respectively. DW-DConv and DW-Conv separately represent $k \times k$ depthwise deconvolution and $k \times k$ depthwise convolution [51], [52] to perform band-wise convolution without band interactions to maintains the originality of spectral domain for further DCR.

In fact, deconvolution is one of the popular upsampling operations in low-level vision tasks [22], [53], [54]. Here, although the adopted depthwise deconvolution–convolution couple only refines the band-wise spatial information without changing the spatial size of $\mathbf{X}_{i,j}$, it is also an upsampling liked transformation to band-wise refine the masked $\mathbf{0}$ vectors by the sampled pixels and further provides more abstract spectral–spatial feature representation compared with $\mathbf{X}_{i,j}^{c1}$.

As illustrated in Fig. 3, for convenience, we uniformly utilize $\mathbf{X}_{i,j}^{c} \in \mathbb{R}^{s\times s\times b}$ to describe the input to construct distance covariance matrix $\mathbf{A}^c \in \mathbb{S}_b^+$ [37], [38]. Specifically, $\mathbf{X}_{i,j}^{c}$ is first reshaped into 2-D matrix $\mathbf{X}^c \in \mathbb{R}^{s^2\times b}$. Then the Euclidean distance of any two bands of $\mathbf{X}^c$ is calculated as

$$\hat{a}_{pq} = \| \mathbf{x}_p - \mathbf{x}_q \|, \quad p, q = 1, \ldots, b \tag{6}$$

where $\mathbf{x}_p \in \mathbb{R}^{s^2}$ and $\mathbf{x}_q \in \mathbb{R}^{s^2}$ represent the pth and qth bands of $\mathbf{X}^c$, respectively, and $\hat{a}_{pq}$ is the corresponding Euclidean distance of $\mathbf{x}_p$ and $\mathbf{x}_q$. Euclidean distance matrix $\hat{A}^c = (\hat{a}_{pq}) \in \mathbb{S}_b^+$ can be further generated. Finally, distance covariance matrix $\mathbf{A}^c$ is calculated as

$$\mathbf{A}^c = \hat{A}^c - \hat{A}^c_{b\cdot} - \hat{A}^c_{\cdot b} + \hat{A}^c_{\cdot\cdot} \tag{7}$$

where $\hat{A}^c_{b\cdot}$, $\hat{A}^c_{\cdot b}$, and $\hat{A}^c_{\cdot\cdot}$ are the row mean, the column mean, and the mean of all the elements of $\hat{A}^c$, respectively.

Thus, we can obtain corresponding distance covariance matrices $\mathbf{A}^{c1}$ and $\mathbf{A}^{c2}$ from channels 1 and 2. By concatenating $\mathbf{A}^{c1}$ and $\mathbf{A}^{c2}$ in the depth direction, we obtain the module output, a dual channel distance covariance representation $\mathbf{A}_0 \in \mathbb{S}_b^+$, which represents low-level spectral–spatial features by spectral-wise construction.

C. Manifold to Euclidean Subspace Learning

The dual channel distance covariance representation $\mathbf{A}_0$ generated from the DC-DCR module is dual channel SPD matrices, which underlie on SPD manifold and require non-Euclidean geometry metrics. As illustrated in Fig. 2, respecting Riemannian geometry of SPD manifold, we implement our M2ESL module to sufficiently exploit the dual channel

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distance covariance representation $\mathbf{A}_0$ in two subspaces, that is, SPD manifold subspace and Euclidean subspace. These two subspace learning and the corresponding Euclidean projection are explained in detail as follows.

1) SPD Manifold Subspace Learning: For the SPD manifold subspace learning, a BiMap-ReEig layer [40] is adopted to produce more compact SPD feature representations and introduce nonlinearity to improve discrimination, that is,

$$\begin{cases} \mathbf{A}_1 = \mathbf{W}_1\mathbf{A}_0\mathbf{W}_1^T \\ \mathbf{A}_1 = \mathbf{U}_1\mathbf{\Sigma}_1\mathbf{U}_1^T \\ \mathbf{A}_2 = \mathbf{U}_1 \max(\epsilon\mathbf{I}, \mathbf{\Sigma}_1)\mathbf{U}_1^T \end{cases} \tag{8}$$

where $\mathbf{A}_1$ and $\mathbf{A}_2$ are the produced intermediate feature maps by the BiMap layer and the ReEig layer, respectively, and $\mathbf{W}_1$, $\mathbf{U}_1$, and $\mathbf{\Sigma}_1$ maintain the same as the corresponding introduction in Sections II-B-II.

Considering that EIG and singular value decomposition (SVD) are GPU-unfriendly and much slower than their CPU counterparts [44], [55], we only employ one BiMap-ReEig layer for SPD manifold subspace learning as a tradeoff between performance and efficiency and adopt GPU-CPU interconversion for EIG in ReEig layer in corresponding programming implementation.

2) Euclidean Projection: In the classic SPD network structure, the LogEig layer is usually employed to project the output feature manifold to a Euclidean subspace. However, the involved EIG is time-consuming even on CPU-based implementation. For one thing, according to [56] and [57], when a positive power approaches 0, the limit of Power-Euclidean metric equals Log-E metric [39]. For another thing, efficient computing of the matrix SQRT [43], [58] has been a hot topic in various vision tasks. Thus, we introduce a Newton–Schulz iteration-based ASQRT method [43], [44] as the Euclidean projecting layer, which only involves GPU-friendly matrix multiplications without matrix inverse.

As depicted in the corresponding flowchart of the ASQRT layer in Fig. 2, to compute the SQRT of SPD feature matrices $\mathbf{A}_2$, the coupled Newton–Schulz iteration is introduced as

$$\begin{aligned} \mathbf{Y}_k &= \frac{1}{2}\mathbf{Y}_{k-1}(3\mathbf{I} - \mathbf{Z}_{k-1}\mathbf{Y}_{k-1}) \\ \mathbf{Z}_k &= \frac{1}{2}(3\mathbf{I} - \mathbf{Z}_{k-1}\mathbf{Y}_{k-1})\mathbf{Z}_{k-1} \end{aligned} \tag{9}$$

where $\mathbf{Y}_0 = \mathbf{A}_F$ and $\mathbf{Z}_0 = \mathbf{I}$ are as the initialization. Moreover, to satisfy the local convergence condition of Newton–Schulz iteration: if $\| \mathbf{A}_F - \mathbf{I} \|_p < 1$, $\mathbf{Y}_k$ and $\mathbf{Z}_k$ quadratically converge to the matrix SQRT $\mathbf{A}_2^{(1/2)}$ and the inverse SQRT $\mathbf{A}_2^{-(1/2)}$, respectively, Frobenius norm (F-norm)-based prenormalization is first performed to guarantee convergence condition before Newton–Schulz iteration. To counteract the change of data magnitude from prenormalization, F-norm-based postcompensation is accordingly performed after Newton–Schulz iteration. These two steps can be described as

$$\mathbf{A}_F = \frac{1}{\| \mathbf{A}_2 \|_F}\mathbf{A}_2, \quad \mathbf{A}_2^{\frac{1}{2}} = \sqrt{\| \mathbf{A}_2 \|_F}\mathbf{Y}_k \tag{10}$$

where $\mathbf{A}_F$ represents F-norm normalized feature matrices, and $\mathbf{A}_2^{(1/2)} \in \mathbb{R}^{b\times b\times 2}$ is the final computed matrix SQRT of $\mathbf{A}_2$, that is, the projected feature maps.

3) Euclidean Subspace Learning: After the ASQRT projecting, we further perform high-level spectral–spatial feature learning in Euclidean subspace. First of all, a batch normalization (BN) [59] layer is applied to $\mathbf{A}_2^{(1/2)}$ instead of adopting a Riemannian BN (RBN) layer [41] in SPD manifold subspace due to the time-consuming computations for EIG, matrix SQRT, and inverse SQRT of the implementation RBN layer, and the corresponding experiment is discussed in Section IV-D. Moreover, the BiMap-ReEig-ASQRT-BN combination plays a vital role in exploiting Riemannian geometry of SPD manifold and high-level spectral–spatial feature, thus we term as a spectral–spatial feature extraction unit (S^2FEU) for convenience.

Furthermore, the 3-D feature map ($b \times b \times 2$) is flattened into a 1-D feature vector ($2b^2$). Then three FC layers are performed further spectral–spatial feature learning in Euclidean subspace. The whole process of Euclidean subspace learning can be expressed as

$$\begin{aligned} \mathbf{A}_3 &= \text{Flatten}\left(\text{BN}\left(\mathbf{A}_2^{\frac{1}{2}}\right)\right) \\ \mathbf{A}_4 &= \text{FC}_1(\text{FC}_0(\mathbf{A}_3)) \\ \mathbf{y}_{i,j} &= \text{Softmax}(\text{FC}_2(\text{Drop}(\text{Sig}(\mathbf{A}_4)))) \end{aligned} \tag{11}$$

where FC_0, FC_1, and FC_2 are the three adopted FC layers, and Sig, Drop, and Softmax are the sigmoid nonlinear function, the dropout regularization avoiding over-fitting, and the softmax function outputting the final class probability vector $\mathbf{y}_{i,j} = [p_1, \ldots, p_K] \in \mathbb{R}^K$. $\mathbf{A}_3$ and $\mathbf{A}_4$ are the intermediate feature maps.

Finally, the classic cross-entropy loss function is employed to train and optimize the proposed AMS-M2EFL framework, which is formulated as

$$\mathcal{L} = -\frac{1}{M}\sum_{m=1}^{M}\sum_{k=1}^{K} \text{In}(y_m = k)\log\left(\hat{y}_m(p_k)\right) \tag{12}$$

where M is the number of samples in a minibatch sample set, y_m is the ground truth of the mth HSI patch sample $\mathbf{P}_m$ in the current minibatch sample set, $\text{In}(y_m = k)$ is the indicator function (if $y_m = k$, $\text{In}(y_m = k)$ is 1; else, $\text{in}(y_m = k)$ is 0), and $\hat{y}_m(p_k)$ is the softmax output probability of $\mathbf{P}_m$ belonging to the kth class.

IV. Experiments

A. Datasets Description and Evaluation Metrics

To comprehensively demonstrate the performance of the proposed AMS-M2ESL framework, we conduct extensive experiments on a class-imbalanced low-resolution dataset IP,[1] a high-resolution dataset (University of Pavia (UP)[1]), and a spatial disjoint high-resolution dataset (University of Houston 13 (UH)).[2]

[1]IP and UP datasets are available from http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes##anomaly_detection

[2]UH dataset is available from https://hyperspectral.ee.uh.edu/?page_id=459

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR IP, UP, AND UH DATASETS

No.	Indian Pines (IP) (5%)			University of Pavia (UP) (1%)			University of Houston (UH) (fixed ratio)		
	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.
1	Alfalfa	3	43	Asphalt	66	6565	Grass-healthy	198	1053
2	Corn-notill	71	1357	Meadows	186	18463	Grass-stressed	190	1064
3	Corn-mintill	41	789	Gravel	20	2079	Grass-synth	192	505
4	Corn	11	226	Trees	30	3034	Tree	188	1056
5	Grass-pasture	24	459	Painted-m-s	13	1332	Soil	186	1056
6	Grass-trees	36	694	Bare-soil	50	4979	Water	182	143
7	Grass-pasture-m	3	25	Bitumen	13	1317	Residentiald	196	1072
8	Hay-windrowed	23	455	Self-block-b	36	3646	Commercial	191	1053
9	Oats	3	17	Shadows	9	938	Road	193	1059
10	Soybean-notill	48	924				Highway	191	1036
11	Soybean-mintill	122	2333				Railway	181	1054
12	Soybean-clean	29	564				Parking-lot1	192	1041
13	Wheat	10	195				Parking-lot2	184	285
14	Woods	63	1202				Tennis-court	181	247
15	Buildings-g-t	19	367				Running-track	187	473
16	Stone-steel-t	4	89						
	Total	510	9739	Total	423	42353	Total	2832	12197

The IP dataset covering the IP agricultural test site in North-western Indiana was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [60] sensor in June 1992. It contains 145 × 145 pixels with 200 bands ranging from 400 to 2500 nm after removing 20 water absorption and low signal-to-noise ratio (SNR) bands. The spatial resolution is 20 m/pixel, and the ground truth is designated into 16 classes of various crops and vegetation with 10 249 labeled pixels. Importantly, the number of samples in some classes of this scene is highly imbalanced.

The UP dataset was captured by reflective optics spectrographic imaging system (ROSIS) [61] during a flight campaign, over Pavia, Northern Italy, 2002. The spatial size of this dataset is 610 × 340 with 42 776 labeled pixels from nine urban classes, and the spatial resolution is 1.3 m/pixel. Each pixel contains 103 spectral bands covering from 430 to 860 nm after removing 12 noisy bands.

The UH dataset was acquired by the Compact Airborne Spectrographic Imager (CASI) 1500 sensor over the University of Houston campus and its neighboring urban area in June 2012. The data contains 349 × 1905 pixels with 144 spectral channels ranging from 364 to 1046 nm and a high spatial resolution of 2.5 m/pixel. In addition, ground-truth reference information was subdivided into spatially disjoint subsets for training and testing, which totally includes 15 mutually exclusive urban classes with 15 029 labeled pixels.

The details of each dataset are uniformly reported in Table I. In detail, about 5% and 1% limited labeled pixels of total labeled pixels per class are randomly selected for training, and the remaining pixels are employed for testing in IP and UP datasets, respectively. For the UH scene, the official fixed dataset division is adopted for training and testing. To quantitatively compare the classification performance of different methods and modules from various aspects, four common evaluation metrics, that is, overall accuracy (OA), average accuracy (AA), kappa coefficient ($\mathcal{K}$), and accuracy of each class (AEC), are employed in the following experiments. Additionally, to quantitatively analyze the model efficiency of different methods, training time (T_{train}) and testing time (T_{test}) are adopted to jointly evaluate the running time of each method, and model parameters (Params) and floating-point operations (FLOPs) are utilized to measure model complexity and computational cost of each method, respectively.

B. Experimental Settings and Model Design

For this work, the hardware environment is a workstation with an Intel Core CPU i9-10900K, 64 GB RAM, and a single Nvidia GeForce RTX 3090 GPU with 24 GB GPU memory. The software environment is Ubuntu 18.04.5 LTS 64 bit, and the programming environment is Python-3.8.5 with the PyTorch-1.10.0 framework. For the AMS-M2ESL training, the Adam [62] optimizer is employed. The batch size and learning rate are set as 256 and 0.001, respectively. The number of training epochs is set as 100 for IP and UP datasets and 40 for the UH dataset.

As discussed above, band-wise de-mean and variance standardization is adopted for all the datasets. Following the standard patch-based processing pattern of the HSIC task, before cropping the 3-D patch for each HSI pixel, the zero padding operation is adopted to pad $(s-1)/2$ zero pixels out of each edge pixel of HSI. In addition, the whole process is repeated ten times to record the average classification accuracy and the standard deviation. For clear comparisons, **boldface** highlights the best results while underline the second in the following suitable table results.

1) Dimension-Reduced Ratio of MNF: In the proposed AMS-M2ESL framework, MNF is employed to reduce dimension in the spectral domain and discard possible noises. Here, this experiment is conducted to select the most proper dimension-reduced ratio r of MNF for the discussed three datasets from a candidate set ranging from 0.1 to 0.45 with an interval of 0.05. As illustrated in Table II, with increasing r, classification results on each dataset exhibit a similar

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TABLE II
CLASSIFICATION RESULTS OF AMS-M2ESL WITH DIFFERENT DIMENSION REDUCED RATIOS OF MNF ON IP, UP, AND UH DATASETS

Data Set	Metric	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45
IP	OA (%)	97.05±0.37	97.53±0.41	98.15±0.39	98.20±0.41	98.38±0.38	98.38±0.40	**98.39±0.46**	98.31±0.46
	AA (%)	97.60±0.29	97.86±0.60	98.60±0.35	98.66±0.35	**98.86±0.26**	98.74±0.33	98.68±0.38	98.39±0.40
	$\mathcal{K} \times 100$	96.64±0.42	97.18±0.46	97.89±0.45	97.95±0.46	98.16±0.43	98.16±0.46	**98.17±0.52**	98.07±0.53
	T_{train}(s)	13.30	18.79	25.46	35.39	42.68	51.13	61.79	77.02
	T_{test}(ms)	4.63	5.90	6.93	8.94	10.54	11.72	13.80	17.13
UP	OA (%)	97.37±0.33	97.82±0.40	98.02±0.26	98.05±0.22	**98.09±0.30**	**98.09±0.30**	97.89±0.32	97.88±0.36
	AA (%)	97.32±0.47	97.48±0.64	97.65±0.39	97.76±0.31	**97.86±0.47**	97.76±0.51	97.65±0.46	97.66±0.45
	$\mathcal{K} \times 100$	96.50±0.43	97.11±0.53	97.38±0.34	97.42±0.30	**97.47±0.39**	97.47±0.40	97.20±0.42	97.19±0.48
	T_{train}(s)	8.58	9.70	12.96	14.82	17.99	21.67	24.43	27.44
	T_{test}(ms)	16.82	19.96	28.21	32.25	40.90	47.43	55.24	61.90
UH	OA (%)	86.21±0.66	87.83±0.46	87.95±0.63	88.50±0.74	**88.82±0.93**	88.63±1.15	88.29±0.83	87.84±0.75
	AA (%)	89.19±0.62	90.84±0.40	91.58±0.60	91.75±0.60	**92.15±0.30**	91.96±0.46	91.73±0.33	91.45±0.48
	$\mathcal{K} \times 100$	85.02±0.72	86.78±0.50	86.91±0.69	87.51±0.80	**87.85±1.01**	87.65±1.25	87.28±0.90	86.79±0.82
	T_{train}(s)	24.59	32.21	41.41	51.41	65.49	79.96	88.67	108.75
	T_{test}(ms)	28.33	38.54	51.69	64.39	82.41	100.93	112.48	137.06

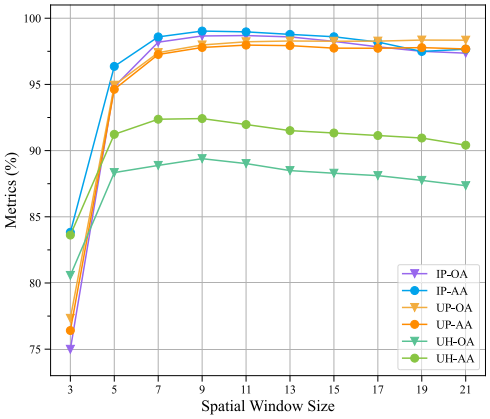


Fig. 4. OAs and AAs of AMS-M2ESL with different spatial window sizes on IP, UP, and UH datasets.

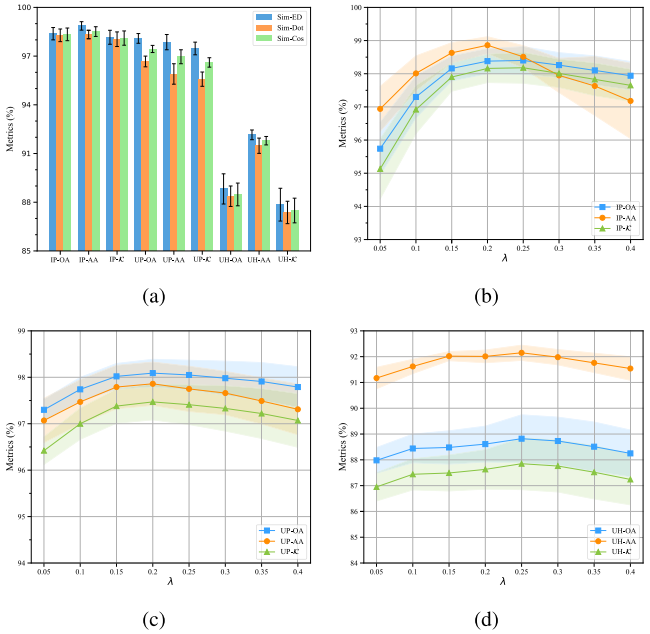


Fig. 5. Setting experiments of the AMIPS module on IP, UP, and UH datasets. (a) Classification results with different similarity measures adopted in the AMIPS module. (b)–(d) Sensitivity analysis of the hyperparameter λ of the AMIPS module. (b) IP dataset. (c) UP dataset. (d) UH dataset.

overall trend, that is, first increase and then decrease. More importantly, when r is set around 0.3, the classification results of the proposed framework reach the peak with moderate running time on all three datasets. In addition, it is clear that the larger dimension-reduced ratio r, the longer the running time. Thus, r is set as 0.3 in the proposed framework on all three datasets.

2) Spatial Window Size: For patch-based HSIC models, the spatial window size of a cropped 3-D patch is a crucial hyperparameter that influences model performance. As shown in Fig. 4, we investigate the proposed AMS-M2ESL framework with a set of spatial window sizes from 3×3 to 21×21 with a step of 2-pixel size on all the discussed datasets. Overall, when spatial window size increases from 3×3 to 9×9, classification results rise rapidly. And the accuracy curves remain stable or even drop when the spatial window size is larger than 9×9, which is more evident in IP and UH scenarios. Therefore, the spatial window size is set as 9×9 in the proposed framework for all the considered datasets in all the following experiments.

3) Similarity Measure in AMIPS Module: In the AMS-M2ESL, we proposed a new sight to alleviate the land-cover inconsistency problem in each patch through the elaborate AMIPS module, which formulates an adaptive mask manner for intrapatch sampling based on the central spectral vector-oriented spatial self-similarity relationships. Here, a corresponding experiment is conducted to study the impact of different similarity measures in the proposed AMIPS module, where SimED, dot product similarity (SimDot), and cosine angle similarity (SimCos) are discussed. As illustrated in Fig. 5(a), the results of SimED are always superior to those of the other two similarity measures on all three datasets, especially on the UP, and the results of SimDot are obviously behind those of the others. Therefore, SimED is employed in the AMIPS module for further AMIPS.

4) Sampling Threshold in the AMIPS Module: In the proposed AMIPS module, we set the sampling threshold t_s as $m_s - \lambda \times s_s$. As long as λ is determined, the further

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adaptive sampling 0-1 mask $\mathbf{M}$ only depends on the central spectral vector-oriented spatial relationships of each patch. This experiment is conducted to explore the impact of λ on the final classification results, where λ ranges from 0.05 to 0.4 with an interval of 0.05. As illustrated in Fig. 5(b)–(d), all the metric results on each dataset are unimodal curves. Moreover, it is clear that classification results reach the peak when λ is around 0.2 for both IP and UP datasets and 0.25 for the UH dataset. Therefore, the best λ values, that is, 0.2, 0.2, and 0.25 are selected for IP, UP, and UH datasets, respectively.

5) Some Other Vital Structure Settings: In particular, we further discuss some other vital structure settings of the AMS-M2ESL. The convolutional kernel size $k \times k$ is set as 5×5 for DW-DConv and DW-Conv in the channel 2 of the DC-DCR module. In the M2ESL module, the channel of the DC-DCR keeps constant at 2 in the SPD manifold subspace learning part and the Euclidean projection part. For the ASQRT layer in the Euclidean projection part, F-norm is adopted in prenormalization and postcompensation instead of trace norm [43], [44] for more stable performance. Moreover, the number of iterations is set as 2 for the coupled Newton–Schulz iteration. The Autograd package in PyTorch is utilized for the backward propagation of the ASQRT layer. For Euclidean subspace learning, the dropout ratio is set as 0.2. The input channels of FC_1 and FC_2 layers are set as 512 and 128 for all the discussed scenes, respectively.

C. Comparison of Classification Performance

To fairly evaluate classification performance and model efficiency of the proposed model, we comprehensively compare the proposed AMS-M2ESL framework with two spectral-based methods, that is, support vector machines (SVMs) with radial basis function (RBF) kernel [63], J-Paly [32],[3] and 1-D-CNN [64],[4] and seven representative state-of-the-art spectral–spatial methods, that is, MCM-CNN [35],[5] SSTN [20],[6] SSSAN [28], SSAtt [11],[7] A^2S^2K-ResNet [12],[8] SSFTT [21],[9] and ASPN [27].[10] In particular, MCM-CNN, SSSAN, and ASPN are reproduced with PyTorch framework according to the corresponding article and source code. The selected compared methods comprehensively cover popular manners and related patterns, including classic classifier, the traditional subspace and manifold learning, 1-D-CNN-based pattern, 2D-CNN-based pattern, 3D-CNN-based pattern, spectral–spatial dual-branch pattern, two-subnetwork structure, second-order pooling, and various attention mechanisms.

For fair comparisons, the adopted classifier in J-Play is the canonical correlation forest (CCF). As for the newly involved scene, for example, the IP dataset for SSSAN, common hyperparameters also keep the same as the unified setting of the original article, and special hyperparameter, for example, preserved components of PCA for the IP scene, is fine-tuned to select the optimal value, and all the existing hyperparameters of compared methods have adopted the recommendation of the corresponding article. Moreover, all the methods take the same experiment settings mentioned above, for example, spatial window size for patch-based methods.

1) Quantitative Accuracy Analysis: Tables III–V list the detailed quantitative classification results of all the discussed methods with limited training samples on IP, UP, and UH datasets. In general, with the limited class-imbalanced condition of the IP dataset, the extremely limited training samples condition of the UP dataset, and the spatial disjoint condition of the UH dataset, the proposed AMS-M2ESL framework always achieves the most remarkable results with clear enhancements compared with any other discussed methods. To be specific, the proposed AMS-M2ESL framework outperforms the suboptimal method by 0.78%, 2.70%, and 0.90 for OA, AA, and $\mathcal{K} \times 100$ on the IP dataset, 1.68%, 2.25%, and 2.24 on the UP dataset, and 2.98%, 0.42%, and 3.24 on the UH dataset. For AEC, AMS-M2ESL obtain the best result on 15 classes of the total 16 classes on the IP dataset, six classes of the total nine classes on the UP dataset, and five classes of the total 15 classes on the UH dataset.

As for the three spectral-based methods, SVM and 1-D-CNN present limited classification results on all three datasets due to the single concentration of the spectral domain without spatial information. For comparisons, J-Paly achieves promising land-cover prediction results on the UH scene with also similar limited classification results on IP and UP datasets, which may benefit from its excellent ability of spectral feature representation. Moreover, MCM-CNN employs MCMs to generate spectral–spatial aggregation for further 2-D-CNN-based pattern feature learning and exhibits clearer improvements on IP and UP scenarios compared with the two spectral-based methods. However, on the UH dataset, MCM-CNN makes the worst results among all the discussed methods, which may be due to the fact that too many scales (15) introduce too many mixed pixels from the sparse land-cover distribution of the UH scene. Then compared to MCM-CNN, SSTN, SSSAN, and SSSAtt achieve further enhancements and show similar performance on all the three datasets by the global spatial correlation mining of transformer, the well-designed spatial and channel attention dense blocks, and the original two-branch spectral–spatial attention structure, respectively.

Furthermore, A^2S^2K-ResNet, SSFTT, and ASPN further improve the classification results on all the discussed datasets by the elaborated 3-D convolution-based model structure, the low-level and high-level feature extraction, and the efficient second-order feature extraction. More importantly, as mentioned above, the proposed AMS-M2ESL framework exhibits much more advanced performance than any other compared methods on all three datasets all the time, which implicitly indicates the importance and significance of the proposed methodology for sufficiently exploring spectral–spatial relationships with discussed different difficult conditions, that is, the limited training samples condition, the class-imbalanced condition, and the spatial disjoint condition.

[3]https://github.com/danfenghong/ECCV2018_J-Play
[4]https://github.com/eecn/Hyperspectral-Classification
[5]https://github.com/henanjun/demo_MCMs
[6]https://github.com/zilongzhong/SSTN
[7]https://github.com/weecology/DeepTreeAttention
[8]https://github.com/suvojit-0x55aa/A2S2K-ResNet
[9]https://github.com/zgr6010/HSI_SSFTT
[10]https://github.com/snowzm/a-spn

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TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET USING 5% LABELED DATA PER CLASS FOR TRAINING

Class	SVM [63]	J-Play [32]	1D-CNN [64]	MCM-CNN [35]	SSTN [20]	SSSAN [28]	SSAtt [11]	A^2S^2K-ResNet [12]	SSFTT [21]	ASPN [27]	AMS-M2ESL
1	42.36±10.46	80.22±15.37	57.69±5.83	87.17±14.33	77.87±39.44	93.98±8.76	96.96±4.70	96.54±4.39	88.03±14.50	99.23±2.31	**100.00±0.00**
2	65.75±2.19	66.85±2.94	74.70±4.59	90.25±2.87	90.17±7.16	92.94±4.51	93.86±3.73	95.23±1.93	90.58±2.18	96.82±1.44	**97.43±1.30**
3	69.58±4.18	67.87±4.64	69.71±6.64	91.84±0.45	91.19±9.12	92.97±5.19	94.58±2.81	95.38±2.08	92.47±2.37	97.24±1.22	**98.86±0.98**
4	50.68±7.63	55.19±8.41	61.61±8.50	90.25±6.58	95.03±6.41	91.67±7.40	93.55±3.62	95.92±4.43	95.91±4.67	96.11±3.56	**98.21±2.25**
5	85.07±4.95	85.06±5.71	84.96±6.09	93.89±3.66	96.86±3.46	98.29±1.65	98.23±1.71	97.53±1.87	98.12±1.44	98.77±0.78	**99.46±0.55**
6	85.17±2.94	84.38±1.92	89.47±3.56	95.26±1.25	98.39±1.67	97.70±2.41	98.26±1.32	98.01±1.06	95.45±2.98	98.73±0.67	**99.70±0.28**
7	72.68±10.11	79.43±12.52	82.38±8.19	82.23±10.01	76.80±17.64	81.71±18.37	88.39±11.03	88.93±14.14	88.66±14.46	91.99±10.55	**100.00±0.00**
8	90.77±1.53	91.68±2.07	93.51±2.06	99.27±0.76	96.21±3.09	97.04±2.27	97.74±2.07	99.55±0.95	95.61±2.34	99.72±0.47	**99.93±0.14**
9	53.21±11.32	54.31±9.37	76.23±10.35	58.12±16.06	70.00±45.83	84.14±12.67	95.84±5.52	79.15±12.01	88.48±14.23	84.11±9.70	**100.00±0.00**
10	69.51±3.29	68.88±4.63	71.52±6.01	93.28±3.13	85.63±12.63	93.40±3.93	91.45±3.59	94.16±2.58	90.39±2.01	94.46±1.79	**95.52±3.15**
11	69.45±1.85	67.03±1.70	73.02±3.58	95.06±1.98	91.54±6.58	94.28±3.54	93.06±2.03	95.85±0.99	91.74±2.56	98.19±0.79	**98.60±0.68**
12	65.83±3.96	69.78±3.38	63.76±3.87	88.61±5.39	89.01±13.31	94.09±5.08	91.77±4.02	95.34±3.03	91.98±2.71	97.48±2.02	**97.67±1.14**
13	89.10±4.19	87.96±3.90	86.54±5.97	89.12±5.91	97.45±4.19	98.99±1.47	99.80±0.41	97.43±2.13	97.56±2.44	97.38±3.00	**100.00±0.00**
14	89.66±2.17	87.30±1.38	88.96±2.83	96.91±2.14	95.80±6.02	96.98±2.19	96.75±1.83	98.73±0.72	96.33±1.98	99.34±0.62	**99.29±0.32**
15	69.62±5.06	64.35±6.13	70.00±10.19	91.98±2.89	91.04±9.13	96.17±2.41	93.91±3.39	98.72±1.70	92.08±3.18	98.19±1.23	**98.73±0.98**
16	**98.92±1.00**	97.74±1.74	94.00±5.13	94.93±4.51	90.47±8.25	94.53±3.16	92.24±4.95	92.03±3.91	90.85±5.63	92.23±5.68	98.35±0.73
OA (%)	74.55±0.66	73.86±1.17	76.94±1.15	93.14±0.91	91.02±2.18	94.56±2.08	94.47±1.02	96.29±0.46	92.93±0.75	97.60±0.44	**98.38±0.38**
AA (%)	72.96±1.21	75.50±1.72	77.38±1.57	89.18±2.05	89.87±3.26	93.43±1.91	94.92±1.03	94.92±1.22	92.85±2.15	96.16±1.17	**98.86±0.26**
$\mathcal{K} \times 100$	70.70±0.74	69.81±1.36	73.56±1.28	92.18±1.04	89.75±2.51	93.78±2.38	93.68±1.17	95.76±0.52	91.92±0.87	97.26±0.51	**98.16±0.43**

TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP DATASET USING 1% LABELED DATA PER CLASS FOR TRAINING

Class	SVM [63]	J-Play [32]	1D-CNN [64]	MCM-CNN [35]	SSTN [20]	SSSAN [28]	SSAtt [11]	A^2S^2K-ResNet [12]	SSFTT [21]	ASPN [27]	AMS-M2ESL
1	81.72±4.46	81.64±1.96	86.73±3.45	82.42±3.02	96.40±2.63	91.95±3.03	95.73±1.56	90.81±2.69	96.41±1.48	89.49±1.78	**97.51±0.88**
2	84.70±1.34	86.93±0.80	86.50±1.06	95.23±1.40	97.05±1.44	92.02±2.42	98.94±0.65	97.71±0.87	98.37±0.47	98.70±0.26	**99.32±0.29**
3	68.64±8.99	78.62±3.59	54.71±7.00	72.66±5.07	91.47±14.34	89.53±6.07	92.42±4.65	86.86±9.58	91.91±4.32	87.39±6.46	**96.06±3.12**
4	93.27±1.63	91.72±3.16	91.68±1.97	85.38±2.91	97.88±1.15	98.48±2.32	**99.54±0.37**	99.38±0.46	98.61±0.88	97.41±0.60	98.86±1.49
5	96.67±1.75	99.49±0.42	93.41±2.54	88.49±2.83	99.10±0.83	98.92±1.15	99.64±0.51	99.50±0.35	99.52±0.50	97.36±0.62	**99.92±0.00**
6	91.05±3.08	81.51±3.54	77.85±5.97	91.13±4.26	**99.44±0.38**	82.44±15.34	98.48±1.33	96.23±2.38	97.28±1.04	99.17±0.33	99.39±0.41
7	64.82±23.62	88.26±5.03	63.60±8.15	79.89±6.47	91.68±16.19	91.91±7.59	93.61±4.25	96.58±4.22	95.30±2.12	96.86±2.65	**99.73±0.36**
8	69.55±4.01	69.57±2.12	68.71±2.82	74.64±1.95	88.55±6.57	85.88±6.03	89.77±3.34	81.26±3.25	85.66±2.52	77.87±3.82	**90.65±2.22**
9	**99.94±0.10**	99.76±0.55	97.50±1.54	62.25±10.15	96.31±4.98	99.09±1.79	99.78±0.23	98.63±1.22	97.42±1.97	94.05±2.61	99.29±0.55
OA (%)	83.31±0.80	84.54±0.48	82.77±0.51	87.70±1.11	95.56±1.62	90.31±3.70	94.41±2.16	94.51±0.73	96.41±0.46	94.55±0.29	**98.09±0.30**
AA (%)	83.37±2.58	86.39±1.08	80.08±1.19	81.34±1.86	95.32±2.56	92.25±2.88	93.40±2.20	94.11±1.22	95.61±0.62	93.15±0.76	**97.86±0.47**
$\mathcal{K} \times 100$	77.10±1.19	79.01±0.65	76.74±0.67	83.68±1.49	94.08±2.15	87.15±4.58	92.57±2.90	92.69±0.97	95.23±0.62	92.75±0.39	**97.47±0.39**

TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UH DATASET USING OFFICIAL FIXED LABELED SAMPLES FOR TRAINING

Class	SVM [63]	J-Play [32]	1D-CNN [64]	MCM-CNN [35]	SSTN [20]	SSSAN [28]	SSAtt [11]	A^2S^2K-ResNet [12]	SSFTT [21]	ASPN [27]	AMS-M2ESL
1	97.85±0.00	97.10±0.00	96.93±1.71	73.13±7.75	94.43±11.92	95.33±2.03	99.48±0.72	**99.52±0.61**	99.34±1.08	99.23±0.57	98.97±0.68
2	**98.65±0.00**	96.24±0.00	95.84±3.29	48.06±4.63	95.65±2.52	91.68±5.27	97.34±2.45	94.92±3.24	96.35±2.47	89.14±3.48	88.93±2.57
3	90.00±0.00	**100.00±0.00**	87.61±12.28	94.75±3.32	79.11±25.67	33.42±23.18	28.52±3.47	99.94±0.13	99.86±6.37	**100.00±0.00**	**100.00±0.00**
4	95.96±0.00	82.76±0.00	95.94±1.39	67.18±2.85	99.02±1.25	97.00±5.09	**99.29±0.43**	84.30±13.25	99.27±4.72	96.23±0.89	99.25±0.24
5	97.05±0.00	99.34±0.00	97.70±1.47	68.32±4.59	98.45±1.16	88.90±6.44	95.00±2.76	99.48±0.70	99.38±6.18	**99.99±0.03**	98.15±1.46
6	10.76±0.00	**100.00±0.00**	11.54±0.82	89.52±1.18	33.52±25.07	54.71±28.28	59.05±15.73	97.92±1.20	98.09±4.38	99.11±1.36	**100.00±0.00**
7	82.83±0.00	71.16±0.00	82.92±4.77	80.68±4.47	86.64±8.99	77.08±6.58	86.89±5.30	82.19±11.57	**92.77±8.76**	81.84±4.78	91.86±2.88
8	71.90±0.00	87.96±0.00	70.89±3.14	53.55±3.88	80.47±9.07	67.19±6.56	84.82±3.15	86.64±2.87	**88.51±2.71**	85.86±0.71	63.94±6.52
9	77.86±0.00	87.42±0.00	76.89±8.04	63.17±4.14	88.47±5.10	87.28±5.62	84.38±3.81	73.00±6.44	**90.32±2.87**	62.59±6.10	79.64±2.32
10	73.58±0.00	80.21±0.00	72.29±4.42	77.53±5.93	73.55±12.05	64.54±6.76	78.06±4.54	65.75±7.49	85.62±2.82	67.49±4.53	**93.26±4.95**
11	71.91±0.00	**90.01±0.00**	78.15±5.61	75.99±5.66	76.65±14.14	87.23±7.89	81.90±6.42	66.78±13.91	46.43±1.31	80.50±5.39	87.32±3.69
12	79.16±0.00	64.84±0.00	75.52±3.10	41.96±7.82	84.02±8.04	65.44±4.71	86.57±1.84	86.60±7.75	**93.04±2.79**	91.71±2.08	92.02±0.88
13	40.00±0.00	53.68±0.00	32.41±4.15	68.35±6.30	89.31±10.83	84.39±4.38	89.57±6.69	90.89±8.36	89.47±0.54	89.46±3.98	**96.46±1.21**
14	90.41±0.00	90.71±0.00	90.49±0.78	97.89±3.24	95.97±6.81	78.66±5.41	99.33±0.92	97.39±3.27	98.80±0.54	**100.00±0.00**	92.51±0.00
15	99.78±0.00	98.94±0.00	99.96±0.09	82.00±7.95	91.10±18.52	98.01±2.23	95.26±1.00	99.12±1.44	98.74±0.54	96.37±1.39	**100.00±0.00**
OA (%)	76.76±0.00	84.80±0.00	75.64±0.89	64.27±0.88	81.61±2.15	74.67±1.03	81.41±0.98	84.13±0.75	85.73±1.41	85.84±0.73	**88.82±0.93**
AA (%)	78.51±0.00	86.69±0.00	77.67±0.92	72.14±0.99	84.42±2.15	78.06±1.92	84.36±1.31	88.30±0.83	91.73±0.76	89.30±0.60	**92.15±0.30**
$\mathcal{K} \times 100$	74.99±0.00	83.51±0.00	73.79±0.96	61.22±0.95	80.13±2.30	72.61±1.11	79.91±1.06	82.77±0.81	84.51±1.53	84.61±0.79	**87.85±1.01**

2) Qualitative Accuracy Analysis: For qualitative evaluation, the classification maps generated by corresponding methods on IP, UP, and UH datasets are visualized from Figs. 6–8. In addition, the three-band false-color composite image and the ground-truth map of each dataset are also, respectively, exhibited for intuitive comparisons. In general, the visualizations of different methods on different datasets are in line with the corresponding classification results recorded in Tables III–V.

Specifically, for the difficult class-imbalanced IP dataset, there are many salt-and-pepper noisy pixels on the classification maps of SVM, J-Play, and 1-D-CNN in visualization due to the lacking of spatial feature extraction. MCM-CNN, SSTN, SSSAN, and SSAtt also exhibit slightly inferior visual results with discontinuous homogeneous regions or large misclassified heterogeneous regions according to the three-band composite image and the ground-truth map. Furthermore, A^2S^2K-ResNet, SSFTT, ASPN, and our proposed AMS-M2ESL framework produce smoother and clearer classification maps. However, some of these four methods fail to maintain stability in all the discussed scenarios. For instance, SSFTT on the IP dataset, ASPN on the UP dataset, and A^2S^2K-ResNet on the UH dataset all exhibit some noisy prediction results, which are obviously contrary to the three-band composite images and ground-truth maps. For comparisons, AMS-M2ESL always yields more accurate and smooth visual results with clear

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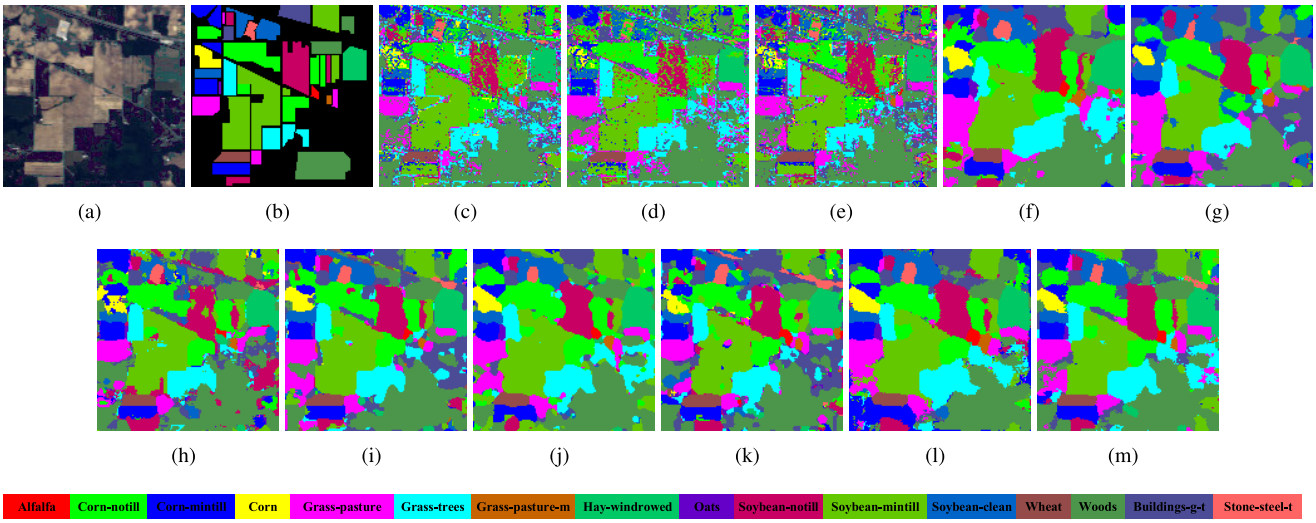


Fig. 6. Classification maps for the IP dataset. (a) Three-band false-color composite image. (b) Ground-truth map. (c) SVM. (d) J-Play. (e) 1-D-CNN. (f) MCM-CNN. (g) SSTN. (h) SSSAN. (i) SSAtt. (j) A^{2}S^{2}K-ResNet. (k) SSFTT. (l) ASPN. (m) AMS-M2ESL.

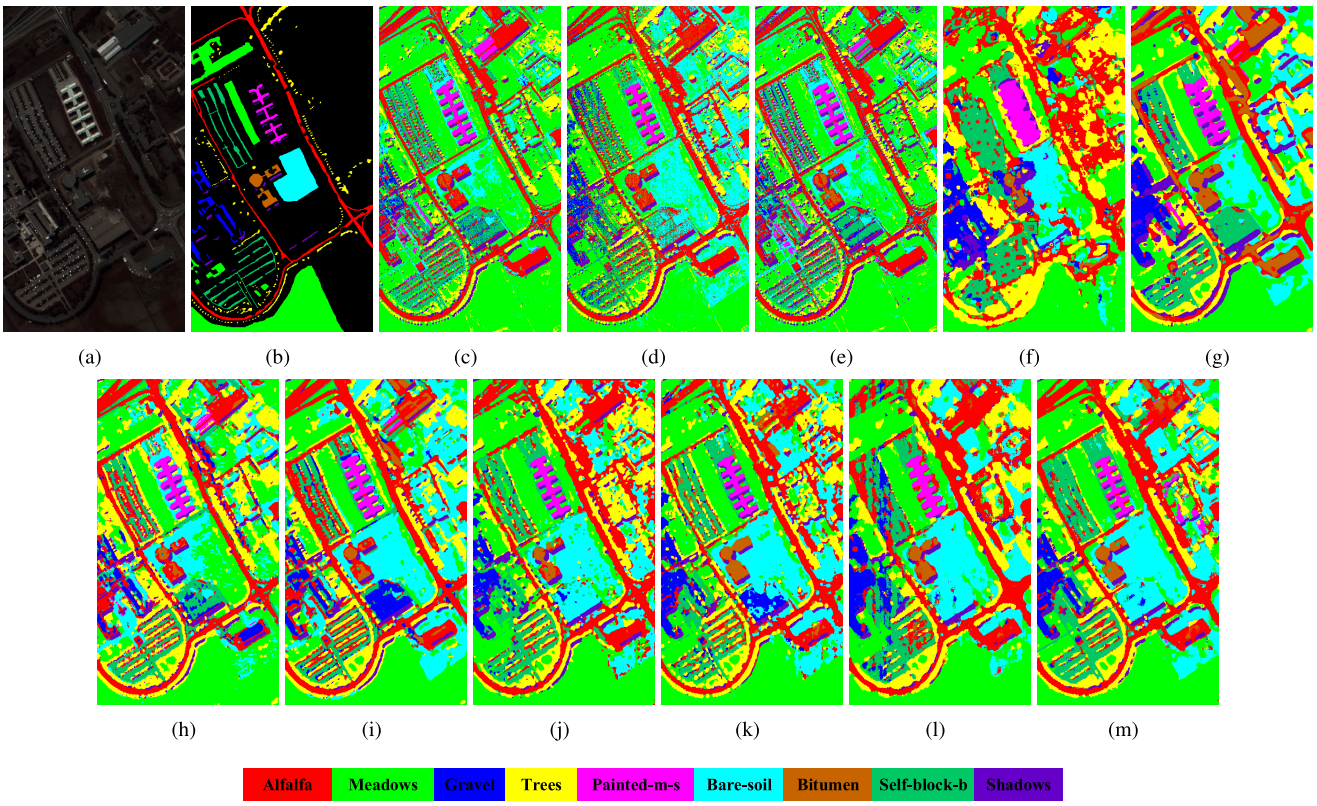


Fig. 7. Classification maps for the UP dataset. (a) Three-band false-color composite image. (b) Ground-truth map. (c) SVM. (d) J-Play. (e) 1-D-CNN. (f) MCM-CNN. (g) SSTN. (h) SSSAN. (i) SSAtt. (j) A^{2}S^{2}K-ResNet. (k) SSFTT. (l) ASPN. (m) AMS-M2ESL.

shapes and boundaries of land covers, which demonstrates its superior spectral–spatial feature extraction ability and stable classification performance even under different difficult conditions of datasets.

3) Efficiency Analysis: Table VI lists the running time, model complexity, and computational cost of different methods for model efficiency comparisons. As mentioned in Section IV-A, T_{train} and T_{test} are adopted to jointly evaluate the running time of each method, and model parameters (Params) and FLOPs are employed to measure model complexity and computational cost of each method, respectively.

In terms of running time, the classic methods, SVM and J-Paly, usually need more time for model testing compared with model training, while deep-learning methods present the opposite time-consuming pattern. Besides, J-Play is slightly time-consuming for training and testing, because of its adopted multiple subspace learning, multicoupled projections, and locally embedding manifold structure.

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TABLE VI
COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP, UP, AND UH DATASETS

Data Set	Metric	SVM [63]	J-Play [32]	1D-CNN [64]	MCM-CNN [35]	SSTN [20]	SSSAN [28]	SSAtt [11]	A^{2}S^{2}K-ResNet [12]	SSFTT [21]	ASPN [27]	AMS-M2ESL
IP	T_{train}(s)	0.03	8.73	2.74	6.02	16.89	53.95	6.93	412.37	5.27	**2.59**	42.68
	T_{test}(ms)	584.62	8281.99	**0.18**	0.92	2.79	17.26	3.76	2.95	1.26	0.36	10.54
	Params (MB)	–	–	0.072	0.184	**0.021**	0.259	0.596	0.371	0.153	0.641	3.772
	FLOPs (MB)	–	–	**0.161**	9.404	1.649	16.188	15.062	171.216	3.661	0.672	4.022
UP	T_{train}(s)	0.03	3.89	2.75	18.02	15.17	44.40	7.46	266.99	5.21	**2.53**	17.99
	T_{test}(ms)	730.36	6328.33	**0.19**	0.93	2.37	16.29	4.69	2.79	1.29	0.34	40.90
	Params (MB)	–	–	0.061	0.654	**0.016**	0.236	0.533	0.221	0.153	0.096	1.057
	FLOPs (MB)	–	–	**0.087**	9.875	1.264	9.945	10.529	87.698	3.660	0.112	1.186
UH	T_{train}(s)	0.11	127.91	5.40	9.87	35.80	103.67	15.51	762.58	11.05	**4.80**	65.49
	T_{test}(ms)	1724.33	22356.97	**0.18**	0.85	2.65	16.45	3.80	2.65	1.32	0.37	82.41
	Params (MB)	–	–	0.066	0.184	**0.018**	0.236	0.563	0.283	0.153	0.316	2.061
	FLOPs (MB)	–	–	**0.112**	10.053	1.427	11.779	12.448	122.498	3.661	0.334	2.244

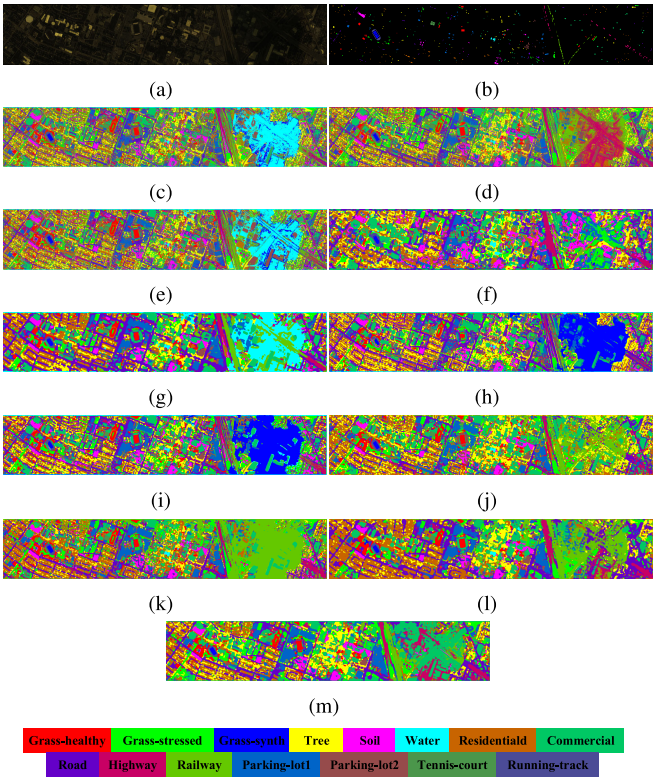


Fig. 8. Classification maps for the UH dataset. (a) Three-band false-color composite image. (b) Ground-truth map. (c) SVM. (d) J-Play. (e) 1-D-CNN. (f) MCM-CNN. (g) SSTN. (h) SSSAN. (i) SSAtt. (j) A^{2}S^{2}K-ResNet. (k) SSFTT. (l) ASPN. (m) AMS-M2ESL.

Moreover, on all three datasets, the well-designed efficient ASPN and the simple 1-D-CNN always lead the top two ranks for model training and testing compared with any other considered deep-learning methods. On the contrary, A^{2}S^{2}K-ResNet and SSSAN always consume more running time among all the deep-learning methods for the time-consuming 3-D convolution backbone and the adopted dense block-based spectral–spatial dual-branch backbone, respectively. Furthermore, the proposed AMS-M2ESL framework shows relatively more time consumption among the discussed deep-learning methods, because the adopted GPU-CPU interconversion and CPU programming are time-consuming in SPD manifold subspace learning of the M2ESL module for respecting Riemannian geometry of SPD manifold. In fact, as a relatively nascent field, it is an open issue for deep manifold learning how to efficiently implement deep manifold structures with limited hardware support respecting Riemannian geometry for manifold data.

As for model parameters, SSTN and 1-D-CNN dominate the top two ranks all the time. Particularly, the Params of SSTN are much fewer than those of other methods owing to the overall efficient self-attention-based model structure. In contrast, AMS-M2ESL exhibits the largest yet acceptable demand for parameter memory on all the datasets due to the employed FC layers in Euclidean subspace learning of the M2ESL module. What is more, 1-D-CNN and ASPN also maintain much lower computational costs than other methods. Considering the excellent classification performances of ASPN on all three datasets, the effectiveness and efficiency of ASPN are comprehensively demonstrated. On the contrary, the FLOPs of A^{2}S^{2}K-ResNet are also the largest, which is also coherent with its huge T_{train} consumption. Moreover, the proposed AMS-M2ESL framework is accompanied by promising computational costs with 4.022, 1.156, and 2.244 MB in FLOPs on IP, UP, and UH datasets.

4) Summary: To sum up, our proposed AMS-M2ESL framework achieves outstanding HSI classification performance compared with the other ten methods in the following three aspects: 1) the remarkable quantitative accuracy in OA, AA, and $\mathcal{K}$ with obvious improvements than the suboptimal method on all the three datasets; 2) the most realistic and stable classification maps with clear shape and boundary of land covers on all the three datasets; and 3) the promising model efficiency for running time, model complexity, and computational cost. Therefore, the excellent performance of the AMS-M2ESL demonstrates the powerful feature representation and extraction ability to cope with different difficult conditions, especially the limited training samples condition, which benefits from the collaborative effort of the AMIPS for the land-cover inconsistency problem, the M2ESL for fully exploring the linear and nonlinear spectral interdependence and extracting high-level spectral–spatial features of the unified DCR respecting Riemannian geometry of SPD manifold.

D. Analysis of S^{2}FEU

As discussed in Section III-C, the S^{2}FEU plays a vital role in exploiting Riemannian geometry of SPD manifold and

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TABLE VII
ANALYSIS OF S^{2}FEU ON IP, UP, AND UH DATASETS, INCLUDING THE BN TEST AND THE ASQRT TEST

Data Set	Test	BiMap	RBN	ReEig	ASQRT	BN	OA (%)	AA (%)	$\mathcal{K} \times 100$	T_{train}(s)	T_{test}(ms)
IP		✓	✗	✓	✓	✓	98.38±0.38	**98.86±0.26**	98.16±0.43	42.68	10.54
	BN	✓	✓	✓	✓	✗	98.35±0.38	98.81±0.24	98.12±0.43	89.86	11.75
		✓	✗	✓	✓	✗	98.27±0.44	98.74±0.30	98.03±0.51	42.31	11.09
	ASQRT	✗	✗	✗	✓	✗	98.13±0.41	98.70±0.27	97.87±0.46	**2.72**	**1.88**
		✓	✗	✓	Log	✓	**98.40±0.42**	98.85±0.27	**98.17±0.48**	79.17	17.45
		✓	✗	✓	SQRT	✓	98.38±0.42	98.83±0.27	98.15±0.47	79.48	17.20
		✓	✗	✓	MPA-Lya	✓	98.38±0.42	98.82±0.28	98.15±0.48	45.65	11.60
UP		✓	✗	✓	✓	✓	98.09±0.30	97.86±0.47	97.47±0.39	17.99	40.90
	BN	✓	✓	✓	✓	✗	**98.18±0.30**	**97.90±0.48**	**97.58±0.39**	39.99	40.03
		✓	✗	✓	✓	✗	97.69±0.34	97.47±0.49	96.93±0.45	17.58	38.76
	ASQRT	✗	✗	✗	✓	✗	97.08±0.32	97.07±0.42	96.11±0.42	**2.23**	**1.98**
		✓	✓	✓	Log	✓	98.10±0.31	97.82±0.49	97.48±0.40	32.00	75.82
		✓	✓	✓	SQRT	✓	98.07±0.29	97.82±0.52	97.44±0.39	31.80	76.04
		✓	✗	✓	MPA-Lya	✓	98.07±0.30	97.82±0.53	97.44±0.39	19.63	43.23
UH		✓	✗	✓	✓	✓	88.82±0.93	92.15±0.30	87.85±1.01	65.49	82.41
	BN	✓	✓	✓	✓	✗	89.16±0.54	**92.32±0.40**	88.23±1.13	141.00	82.64
		✓	✗	✓	✓	✗	88.30±1.19	91.20±0.70	87.29±1.29	64.43	85.65
	ASQRT	✗	✗	✗	✓	✗	86.66±0.54	89.90±0.50	85.51±0.58	**3.70**	**2.22**
		✓	✗	✓	Log	✓	88.88±0.79	91.88±0.33	87.93±0.86	115.70	151.49
		✓	✗	✓	SQRT	✓	89.17±0.70	91.98±0.39	88.23±0.76	119.24	156.02
		✓	✗	✓	MPA-Lya	✓	**89.19±0.63**	91.95±0.37	**88.26±0.68**	69.30	86.42

high-level spectral–spatial features. In this section, we conduct two break-down ablation tests to analyze the setting of the BN layer in S^{2}FEU and the effectiveness of the adopted ASQRT layer. For clear comparisons, the standard classification results of AMS-M2ESL are also recorded.

1) BN Test: For this test, we compare the BiMap-RBN-ReEig-ASQRT and BiMap-ReEig-ASQRT combinations on all three datasets. Refer to [41] for the detailed theoretical justifications and properties of the RBN layer. As illustrated in Table VII, for one thing, when the RBN layer is employed instead of the classic BN layer, there are a slight performance drop on the IP scene and subtle performance improvements on UP and UH scenes. However, on IP, UP, and UH datasets, the T_{train} consumption increases more than one time, that is, from 42.68 to 89.86 s, from 17.99 to 39.99 s, and from 65.49 to 141.00 s, respectively, which is due to the involved many time-consuming computations for EIG, matrix SQRT, and inverse SQRT in the implementation of RBN layer. For another thing, for S^{2}FEU without the BN layer, the classification performance of the AMS-M2ESL obviously drops on all the datasets. Thus, in the proposed AMS-M2ESL framework, we select the BiMap-ReEig-ASQRT-BN combination in the proposed M2ESL module to mine high-level spectral–spatial features respecting Riemannian geometry of the SPD manifold.

2) ASQRT Test: In this test, we further analyze the effort of the adopted ASQRT layer for Euclidean projection from SPD manifold subspace to Euclidean subspace. First, as shown in Table VII, we only employ the ASQRT layer of the S^{2}FEU. The classification results on all the datasets descend a lot, especially more than 1% and 2% on UP and UH datasets. At the same time, the consumed time greatly decreases, which is less than 4 s on all three datasets. Thus, as discussed above, the adopted GPU–CPU interconversion and CPU programming consume most part of the running time for the proposed AMS-M2ESL framework.

Furthermore, we replace the adopted ASQRT layer with two classic non-Euclidean metrics, that is, matrix logarithm (Log) [39] and matrix SQRT [56], [57], and a popular differentiable matrix SQRT method, that is, Matrix Padé Approximants and iterative Lyapunov solver (MPA-Lya) [58]. As recorded in Table VII, Log and SQRT achieve similar classification results with ASQRT on all the datasets, but require nearly two times the demand for running time. Meanwhile, MPA-Lya presents a similar performance to ASQRT with a little more time consuming due to solving the linear system on the CPU. Above all, the adopted ASQRT maintains excellent classification results and relatively efficient model efficiency for Euclidean projection in the proposed M2ESL module.

E. Ablation Study

In this section, we conduct ablation experiments on IP, UP, and UH datasets to further explore and validate the contributions of different modules of the proposed AMS-M2ESL framework, that is, MNF, the AMIPS module, the DC-DCR module, and the S^{2}FEU of the M2ESL module. The corresponding results are listed in Table VIII.

1) Ablation for MNF: In the proposed AMS-M2ESL framework, MNF is employed to reduce spectral dimension and discard possible noises. Here, we replace MNF with the classic PCA of the corresponding same component number to verify the effect of MNF. As shown in Table VIII, MNF brings obvious performance gains over PCA-embedded

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TABLE VIII
ABLATION STUDY ON IP, UP, AND UH DATASETS

Data Set	MNF	AMIPS	DCR	S^2FEU	OA (%)	AA (%)	$\mathcal{K} \times 100$	T_{train}(s)	T_{test}(ms)
IP	✓	✓	✓	✓	**98.38±0.38**	**98.86±0.26**	**98.16±0.43**	42.68	10.54
	PCA	✓	✓	✓	97.51±0.46	97.52±0.86	97.16±0.52	43.13	10.37
	✓	✗	✓	✓	97.78±0.45	96.75±1.10	97.46±0.51	42.56	10.07
	✓	✓	CR	✓	98.06±0.40	98.62±0.31	97.79±0.45	41.64	10.12
	✓	✓	✓	✗	98.06±0.43	98.63±0.29	97.79±0.49	**1.80**	**1.26**
UP	✓	✓	✓	✓	**98.09±0.30**	**97.86±0.47**	**97.47±0.39**	17.99	40.90
	PCA	✓	✓	✓	97.86±0.34	97.51±0.35	97.16±0.45	17.80	40.25
	✓	✗	✓	✓	96.84±0.52	95.95±0.62	95.80±0.69	17.70	39.75
	✓	✓	CR	✓	97.76±0.32	97.30±0.57	97.03±0.43	16.75	37.81
	✓	✓	✓	✗	96.93±0.25	96.78±0.39	95.91±0.33	**1.86**	**1.20**
UH	✓	✓	✓	✓	**88.82±0.93**	**92.15±0.30**	**87.85±1.01**	65.49	82.41
	PCA	✓	✓	✓	86.52±0.38	91.08±0.74	85.36±0.41	65.24	80.75
	✓	✗	✓	✓	88.42±0.90	91.83±0.48	87.43±0.98	63.04	79.33
	✓	✓	CR	✓	84.45±0.53	88.67±0.50	83.12±0.58	61.58	79.13
	✓	✓	✓	✗	86.33±0.74	89.64±0.88	85.15±0.81	**2.71**	**1.21**

AMS-M2ESL, especially more than 1% gains on the UH scene.

2) Ablation for AMIPS: As a new insight to alleviate the spatial land-cover inconsistency problem, the proposed AMIPS module formulates an adaptive mask manner for intrapatch sampling based on the central spectral vector-oriented spatial relationships. Thus, we test the AMS-M2ESL framework without the AMIPS module in Table VIII. It is clear that there are varying degrees of classification performance degradation without the AMIPS module on all the datasets, for example, 0.60%, 2.11%, and 0.70 on the IP dataset and 1.25%, 1.91%, and 1.67 on UP dataset for OA, AA, and $\mathcal{K} \times 100$, respectively.

3) Ablation for DC-DCR: In the proposed DC-DCR module, based on the distance covariance descriptor, we model unified spectral–spatial feature representations and explore spectral–spatial relationships, especially linear and nonlinear interdependence in the spectral domain. To validate the key role of distance covariance descriptor, we replace the DCR with classic covariance representation (CR) in the DC-DCR module. From Table VIII, although we maintain the dual channel representation structure except for the displacing of distance covariance descriptor, clear decreases occur on all the scenes. More importantly, there are about 4.37%, 3.48%, and 4.73 performance descents for OA, AA, and $\mathcal{K} \times 100$ on the UH dataset, which explicitly indicates the importance of modeling and exploiting the nonlinear interdependence of HSI.

4) Ablation for $\mathcal{S}^2$FEU: At last, we evaluate the classification performance of the proposed AMS-M2ESL framework without the total $\mathcal{S}^2$FEU in the M2ESL module to explore the corresponding improvements on all the datasets. As illustrated in Table VIII, performance gains produced by $\mathcal{S}^2$FEU are evident on all the discussed scenarios, especially on UP and UH datasets, where are 1.16%, 1.08%, and 1.56 as well as 2.49%, 2.51%, and 2.70 for OA, AA, and $\mathcal{K} \times 100$, respectively. Thus, it indicates the importance of exploiting the Riemannian geometry of the SPD manifold for the dual-channel DCR.

F. Impact of Training Sample Proportion

In order to further explore the robustness of all the discussed eleven different methods, we also investigate the classification performances of different methods with different proportions of labeled samples for model training on IP, UP, and UH datasets. It is noteworthy that the UH dataset is also random sampling per class as IP and UP datasets instead of official spatially disjoint subdividing. Concretely, 0.5%, 1.0%, 2.5%, 5.0%, and 7.5% labeled samples per class are randomly selected as training sets for the relatively small sample dataset, IP, and 0.25%, 0.5%, 1.0%, 2.5%, and 5.0% for the two relatively large sample dataset, UP and UH. Besides, the extremely limited labeled samples conditions are considered on all the datasets, that is, 0.5% and 1.0% on the IP dataset and 0.25% and 0.5% on the UP and UH datasets. Fig. 9 illustrates the changing trends of OA, AA, and $\mathcal{K}$ as the proportion of training samples increases on all the datasets.

In total, the overall trend of the curve for each method is rising with the percentage of training samples increasing. To be specific, SVM and 1-D-CNN still show limited performances with different amounts of training samples on all the datasets. Importantly, unlike the similar weak performance of J-Play on IP and UP scenes, the curves of J-Play on all the metrics of the UH dataset show promising classification performance, which is also coherent with its classification result in Table V. With the growth of training samples, MCM-CNN exhibits rapid improvements in classification performances on all the datasets. Furthermore, SSTN, SSSAN, and A^2S^2K-ResNet present close growing trends and classification results. What is more, on the IP dataset, AMS-M2ESL and ASPN always dominate the top two ranks at each considered proportion of training samples. For the UP dataset, AMS-M2ESL, SSFTT, and SSAtt achieve distinguished classification results all the time. The curves of AMS-M2ESL, ASPN, and SSFTT exhibit excellent classification performances for all the considered training set settings on the UH dataset. Moreover, compared with ASPN, SSFTT, and SSAtt, the proposed AMS-M2ESL framework always leads

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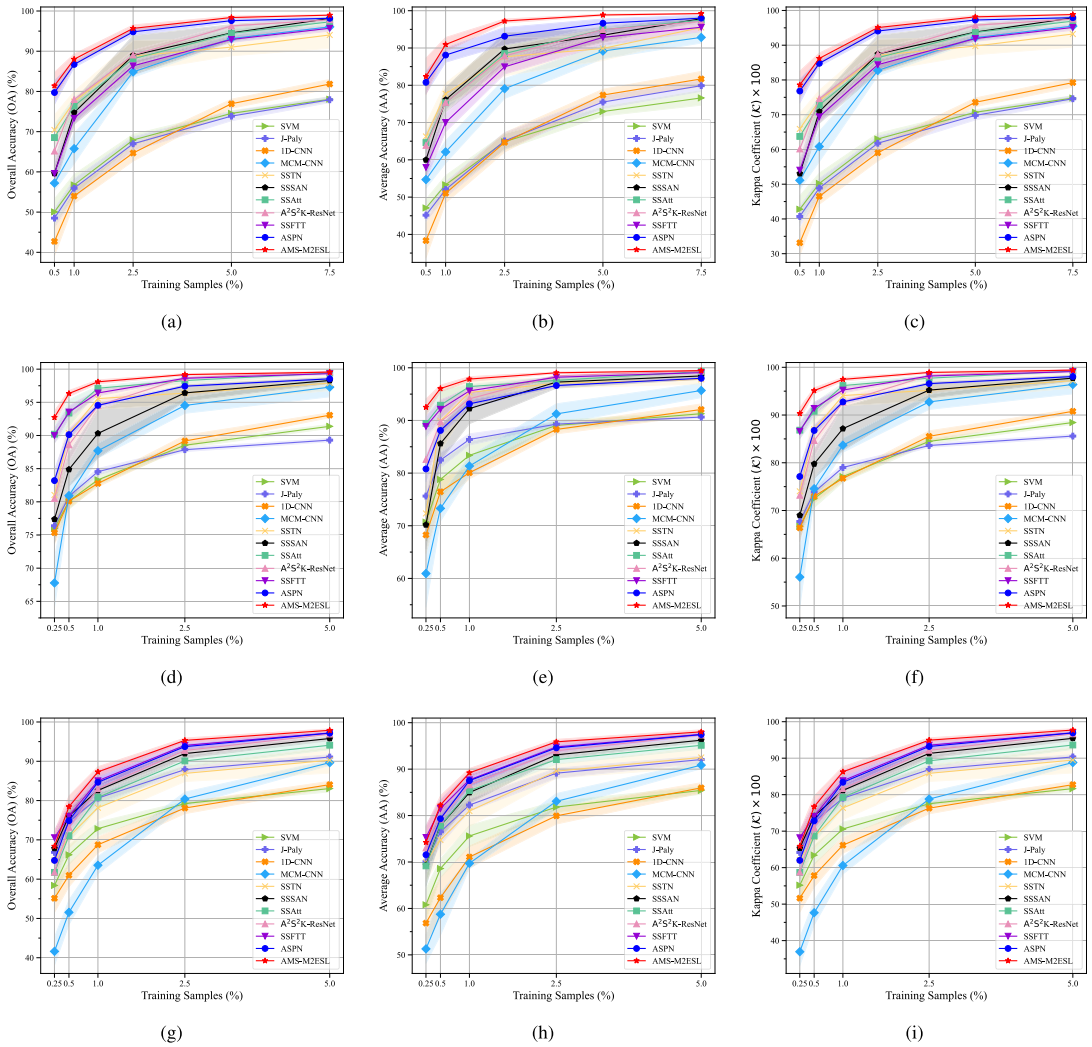


Fig. 9. Classification results of different methods on IP and UP datasets with varying amounts of training samples. (a) OAs on IP dataset. (b) AAs on IP dataset. (c) $\mathcal{K}$s on IP dataset. (d) OAs on UP dataset. (e) AAs on UP dataset. (f) $\mathcal{K}$s on UP dataset. (g) OAs on UH dataset. (h) AAs on UH dataset. (i) $\mathcal{K}$s on UH dataset.

with evident enhancements at different percentages of training samples on all the datasets, which validates the powerful ability of the proposed framework even with extremely limited labeled samples to sufficiently mine spectral–spatial relationships, represent low-level features, and exploit high-level features.

G. Feature Visualizations by t-SNE

To intuitively evaluate the feature representation ability of our framework, in this section, we further utilize t-distributed stochastic neighbor embedding (t-SNE) [65] to visualize the feature distribution learned by three competitive methods, that is, A^{2}S^{2}K-ResNet, ASPN, and SSFTT, with the proposed AMS-M2ESL framework. As shown in Fig. 10, all the test set samples of different datasets are employed for t-SNE visualization. Because amounts of test set samples on different scenes are distinct, the visualization results on different scenes are also different. It is clear that the proposed AMS-M2ESL framework has higher distinguishability with fewer misclassified samples than the other three methods, especially on the IP dataset [Fig. 10(d)]. In addition, the visual results of ASPN are slightly inferior with scattered class clusters [Fig. 10(g) and (l)].

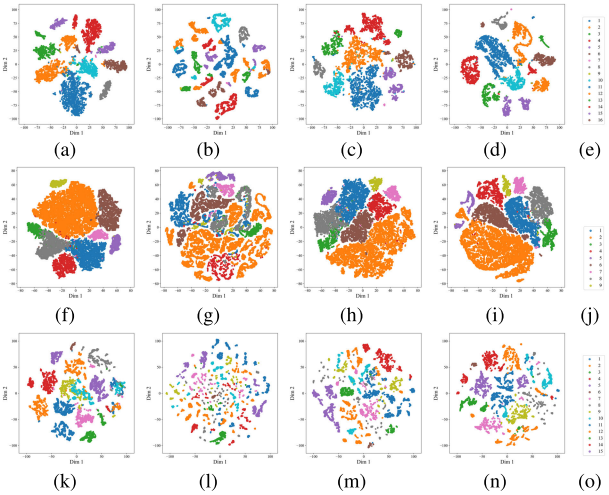


Fig. 10. t-SNE visualization results of A^{2}S^{2}K-ResNet, ASPN, SSFTT, and AMS-M2ESL in sequence on IP, UP, and UH datasets. (a)–(e), (f)–(j), and (k)–(o) Visualization results of different methods on IP, UP, and UH datasets with the corresponding legend, respectively.

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V. Conclusion

For the HSIC task, the spatial land-cover inconsistency problem and complex spectral nonlinear interdependence of HSIs are two intractable obstacles to fully exploring spectral–spatial relationships. In this article, we propose an AMS-M2ESL framework with limited training samples for HSIC. Specifically, we first formulate the AMIPS module for intrapatch sampling in an adaptive mask manner as a new insight to alleviate the discussed spatial land-cover inconsistency problem. Second, based on the distance covariance descriptor, we design the DC-DCR module for modeling unified spectral–spatial feature representations and exploring linear and nonlinear spectral interdependence. Furthermore, we implement the M2ESL module respecting the Riemannian geometry of the SPD manifold for high-level spectral–spatial feature learning. In addition, the ASQRT layer is introduced for Euclidean projecting between the SPD manifold subspace and Euclidean subspace. Extensive experimental results demonstrate that the three proposed modules of AMS-M2ESL greatly improve the ability of mining spectral–spatial relationships and extract discriminant spectral–spatial features to achieve outstanding classification performances with limited labeled samples on three popular HSI datasets compared with other state-of-the-art methods.

In the future, we will further optimize the efficiency of SPD manifold subspace learning with deep manifold learning. What is more, we will still pay more attention to sufficiently exploring and exploiting the spectral–spatial relationships.

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References

[1] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. A. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Sep. 2012.

[2] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[3] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[4] C.-I. Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.

[5] A. Martínez-Usó, F. Pla, J. M. Sotoca, and P. García-Sevilla, "Clustering-based hyperspectral band selection using information measures," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4158–4171, Dec. 2007.

[6] S. Prasad and L. M. Bruce, "Limitations of principal components analysis for hyperspectral target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 625–629, Oct. 2008.

[7] A. A. Green, M. Berman, P. Switzer, and M. D. Craig, "A transformation for ordering multispectral data in terms of image quality with implications for noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. GRS-26, no. 1, pp. 65–74, Jan. 1988.

[8] M. Wang, J. Yu, L. Niu, and W. Sun, "Unsupervised feature extraction for hyperspectral images using combined low rank representation and locally linear embedding," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2017, pp. 1428–1431.

[9] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[10] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[11] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[12] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[13] C. Zhao, W. Zhu, and S. Feng, "Superpixel guided deformable convolution network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3838–3851, 2022.

[14] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2020.

[15] J. Jiang, J. Ma, and X. Liu, "Multilayer spectral–spatial graphs for label noisy robust hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 839–852, Feb. 2022.

[16] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[17] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2016.

[18] T. Song, Y. Wang, C. Gao, H. Chen, and J. Li, "MSLAN: A two-branch multidirectional spectral–spatial LSTM attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528814.

[19] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[20] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715.

[21] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[22] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[23] J. Jiang, C. Wang, X. Liu, K. Jiang, and J. Ma, "From less to more: Spectral splitting and aggregation network for hyperspectral face super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 267–276.

[24] N. Audebert, B. L. Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[25] H. Sun, X. Zheng, and X. Lu, "A supervised segmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 2810–2825, 2021.

[26] T.-Y. Lin, A. RoyChowdhury, and S. Maji, "Bilinear CNN models for fine-grained visual recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1449–1457.

[27] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[28] X. Zhang et al., "Spectral–spatial self-attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512115.

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[29] C. M. Bachmann, T. L. Ainsworth, and R. A. Fusina, "Exploiting manifold geometry in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 441–454, Mar. 2005.

[30] D. Lunga and O. Ersoy, "Spherical stochastic neighbor embedding of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 857–871, Feb. 2013.

[31] D. Lunga, S. Prasad, M. M. Crawford, and O. Ersoy, "Manifold-learning-based feature extraction for classification of hyperspectral data: A review of advances in manifold learning," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 55–66, Jan. 2014.

[32] D. Hong, N. Yokoya, J. Xu, and X. Zhu, "Joint & progressive learning from high-dimensional data for multi-label classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 469–484.

[33] Z. Li, H. Huang, Z. Zhang, and Y. Pan, "Manifold learning-based semisupervised neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508712.

[34] L. Fang, N. He, S. Li, A. J. Plaza, and J. Plaza, "A new spatial–spectral feature extraction method for hyperspectral images using local covariance matrix representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3534–3546, Mar. 2018.

[35] N. He et al., "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.

[36] S. Zhang, T. Lu, S. Li, and W. Fu, "Superpixel-based Brownian descriptor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520112.

[37] G. J. Székely, M. L. Rizzo, and N. K. Bakirov, "Measuring and testing dependence by correlation of distances," *Ann. Statist.*, vol. 35, no. 6, pp. 2769–2794, 2007.

[38] G. J. Székely and M. L. Rizzo, "Brownian distance covariance," *Ann. Appl. Statist.*, vol. 3, no. 4, pp. 1236–1265, 2009.

[39] V. Arsigny, P. Fillard, X. Pennec, and N. Ayache, "Geometric means in a novel vector space structure on symmetric positive-definite matrices," *SIAM J. Matrix Anal. Appl.*, vol. 29, no. 1, pp. 328–347, 2007.

[40] Z. Huang and L. Van Gool, "A Riemannian network for SPD matrix learning," in *Proc. 31st AAAI Conf. Artif. Intell.*, 2017, pp. 1–7.

[41] D. Brooks, O. Schwander, F. Barbaresco, J.-Y. Schneider, and M. Cord, "Riemannian batch normalization for SPD neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.

[42] R. Wang, X.-J. Wu, and J. Kittler, "SymNet: A simple symmetric positive definite manifold deep learning method for image set classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 5, pp. 2208–2222, May 2022.

[43] P. Li, J. Xie, Q. Wang, and Z. Gao, "Towards faster training of global covariance pooling networks by iterative matrix square root normalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 947–955.

[44] Q. Wang, J. Xie, W. Zuo, L. Zhang, and P. Li, "Deep CNNS meet global covariance pooling: Better representation and generalization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2582–2597, Aug. 2020.

[45] R. Wang, H. Guo, L. S. Davis, and Q. Dai, "Covariance discriminative learning: A natural and efficient approach to image set classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2496–2503.

[46] S. Bak, M. S. Biagio, R. Kumar, V. Murino, and F. Bremond, "Exploiting feature correlations by Brownian statistics for people detection and recognition," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 47, no. 9, pp. 2538–2549, Sep. 2017.

[47] J. Xie, F. Long, J. Lv, Q. Wang, and P. Li, "Joint distribution matters: Deep Brownian distance covariance for few-shot classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, May 2022, pp. 7972–7981.

[48] V. Nair and G. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 807–814.

[49] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[50] N. Lang, N. Kalischek, J. Armston, K. Schindler, R. Dubayah, and J. D. Wegner, "Global canopy height regression and uncertainty estimation from GEDI LiDAR waveforms with deep ensembles," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112760.

[51] V. Dumoulin and F. Visin, "A guide to convolution arithmetic for deep learning," 2016, *arXiv:1603.07285*.

[52] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.

[53] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[54] Y. Cai et al., "Mask-guided spectral-wise transformer for efficient hyperspectral image reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17502–17511.

[55] C. Ionescu, O. Vantzos, and C. Sminchisescu, "Matrix backpropagation for deep networks with structured layers," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2965–2973.

[56] I. L. Dryden, A. Koloydenko, and D. Zhou, "Non-Euclidean statistics for covariance matrices, with applications to diffusion tensor imaging," *Ann. Appl. Statist.*, vol. 3, no. 3, pp. 1102–1123, 2009.

[57] P. Li, J. Xie, Q. Wang, and W. Zuo, "Is second-order information helpful for large-scale visual recognition?" in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2070–2078.

[58] Y. Song, N. Sebe, and W. Wang, "Fast differentiable matrix square root," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 182–201.

[59] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[60] G. Vane, R. O. Green, T. G. Chrien, H. T. Enmark, E. G. Hansen, and W. M. Porter, "The airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 127–143, 1993.

[61] B. Kunkel, F. Blechinger, R. Lutz, R. Doerffer, and H. van der Piepen, "ROSIS (reflective optics system imaging spectrometer)—A candidate instrument for polar platform missions," *Optoelectron. Technol. Remote Sens. Space*, vol. 868, pp. 134–141, May 1988.

[62] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[63] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[64] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Aug. 2015.

[65] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 1–27, Nov. 2008.



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